Department of Computer Science



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Integrated Database and Large Language Modelling for Clinical Support

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STATEMENT OF ETHICS

Legal. The clinical guideline component reuses text from NICE guideline NG101 [44] under its open licence, which permits text and data mining and use with an AI model, with the source clearly referenced. No other legal issue, including data protection, arises beyond what is addressed under Ethical below.

Social. MDT-based cancer care depends on specialist availability that is unevenly distributed internationally, which motivates the project. The project makes no claim to address that disparity, only to evaluate a technical feasibility question within it; Section 1.2 states explicitly that the system is not proposed as a replacement for the MDT.

Ethical. The project uses retrospective, de-identified clinical records from three sources (KATH, SGMC and SEER); no data was collected from patients and no attempt was made to re-identify anyone. KATH and SGMC extracts carried no direct identifier, and SEER is a publicly available dataset de-identified under US federal statistical disclosure requirements. The retrieval-augmented component raises a further consideration, addressed in Section 5.7: it is evaluated only as a proof-of-concept, its outputs include categorised failure modes such as fabricated content, and it is not proposed as a validated or deployable clinical decision aid.

Professional. Source material is acknowledged in the department's required referencing style. Findings that did not support the project's hypotheses, including a withdrawn agreement figure and a corrected survival analysis, are reported rather than omitted, consistent with the standard of honest reporting regardless of outcome.

Commercial. The project has no commercial sponsor, proposes no commercial application, and involves no conflict of interest with any organisation whose products or guidelines are discussed.

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Executive Summary

Multidisciplinary team (MDT) meetings are the standard of care in breast cancer management, but they depend on the simultaneous availability of several scarce specialists, a dependency that becomes a structural bottleneck where specialist coverage is limited. Machine learning has been proposed to support MDT decision-making, yet almost all existing work is developed and evaluated inside single, well-resourced institutions, and little is known about whether such models transfer to the very different data environments of the hospitals that stand to benefit most.

This dissertation designs, implements and evaluates an end-to-end supervised learning pipeline that predicts breast cancer MDT treatment recommendations and remains applicable across institutions that record data differently. The work follows the Cross-Industry Standard Process for Data Mining (CRISP-DM), with harmonisation preceding modelling and the test partition held untouched until final evaluation. Three datasets are used: 5,000 MDT records from Komfo Anokye Teaching Hospital (KATH) in Ghana, 1,000 records from the Sweden Ghana Medical Centre (SGMC), and 4,024 records from the United States SEER programme, chosen as the most demanding available combination for testing cross-institutional transfer, spanning two hospitals with differing schemas and a national registry with a different coding standard and population entirely.

A configurable harmonisation layer reconciles both column names and value encodings across these sources in two ordered stages: column mapping aligns differently named fields describing the same clinical concept, and value mapping reconciles differing vocabularies for the same field, such as HER2 status recorded as a clinical category at one site and a raw immunohistochemistry score at another. Three nested feature sets of five, fifteen and thirty-two variables represent differing levels of data availability, reflecting what a hospital at each level of data-capture maturity could realistically supply. The prediction target is reported in two forms, the original fifteen MDT categories and six clinically coherent treatment groups, and a controlled re-analysis, holding cohort and partition fixed across configurations, separates the effect of feature richness from the effect of classifier choice.

As an extension to the core pipeline, a retrieval-augmented generation (RAG) layer grounds each prediction in an explanation drawn from a licensed clinical guideline (NICE NG101), retrieving relevant passages and constraining a language model to explain the recommendation using only that retrieved text. The classifier remains the sole decision-maker; the RAG layer's role is confined to producing a citation-grounded rationale, evaluated for concordance with guideline content

and faithfulness to the retrieved passages, including a documented redesign after an initial evaluation identified a fabrication pattern.

On the KATH held-out test set, the fifteen-class task achieved 75.29% accuracy (macro-F1 0.566) with the full feature set, and the six-group task achieved 86.73% accuracy (macro-F1 0.742), confirmed at 87.60% (SD 0.94%) under five-fold cross-validation. Transferred without retraining to SGMC, the harmonised model produced coherent, distributed predictions on 992 patients. Applied across continents to SEER, the model produced stage-coherent recommendations. A controlled re-analysis, holding cohort and partition fixed, establishes the principal empirical finding: predictive performance is governed more by the richness of recorded data than by the choice of classifier, with the effect of feature richness between 2.4 and 4.5 times that of the algorithm.

The analysis also reports limitations candidly: one recommendation category is never recovered because it is an unlearnable residual rather than a treatment decision, a cross-hospital agreement figure is withdrawn once trivial predictors are shown to match it, and a cross-continental survival association is shown to be 93% attributable to disease stage. The work is offered as an engineering and feasibility study rather than a clinically validated decision aid.

1 Introduction

1.1 Background and Motivation

Cancer is among the leading causes of death worldwide, and its management follows a defined pathway spanning screening, diagnosis, staging and treatment. A pivotal stage is the Multidisciplinary Team (MDT) meeting, or tumour board, in which oncologists, surgeons, radiologists, pathologists and specialist nurses jointly review individual cases and agree a treatment recommendation [1]. MDT working is regarded as the standard of care in oncology because it brings several specialisms to bear on each decision, improving the consistency and quality of the treatment plan. That strength is also a constraint: MDT meetings depend on the simultaneous availability of several scarce specialists. In breast cancer, the most commonly diagnosed cancer in women worldwide and responsible for an estimated 670,000 deaths in 2022 [8,9], the volume of cases can outstrip the capacity of MDT meetings to review them thoroughly, creating a bottleneck felt most acutely in lower-resource systems where specialist coverage is limited and survival outcomes remain poorer.

This bottleneck motivates interest in machine learning (ML) as a means of supporting, rather than replacing, MDT decision-making. Applied to structured clinical data, ML can propose an evidence-consistent recommendation for the team to review, or prioritise cases where specialist time is scarce. A recent systematic review and meta-analysis of AI-based decision-support systems for tumour diseases confirms both the rapid growth of this field and the wide variability in reported performance [7]. Yet almost all of this work has been developed and evaluated within single, well-resourced institutions in high-income countries, and whether such systems transfer to the very different data environments of hospitals in lower-resource settings remains largely untested. It is this question of cross-institutional and cross-continental generalisation that motivates the project.

Realising such generalisation raises a further problem: data heterogeneity. Different centres record the same clinical information differently. HER2, a protein whose level helps determine breast cancer treatment, may be recorded at one centre as a simple category such as positive or negative and at another as a graded score, under entirely different field names. The clinical meaning is identical, but the labels and formats differ. Without a systematic means of reconciling these differences, a model trained at one hospital cannot be applied at another. This dissertation therefore treats data harmonisation as a first-class design concern rather than a preprocessing afterthought.

1.2 Problem Statement

Existing ML approaches to MDT decision support are trained and evaluated at a single institution, incorporate no mechanism for reconciling schema differences across hospitals, and are not tested on an external population. As a result, it is currently unknown whether a supervised pipeline trained on real-world MDT records from one hospital can (i) predict MDT treatment recommendations at clinically useful accuracy, (ii) generalise to a second hospital whose data schema differs substantially, and (iii) transfer to a dataset drawn from an entirely different population and coding standard. The system predicts the recommendation recorded by the MDT, the decision actually taken rather than the clinically optimal one, and is not proposed as a replacement for the MDT: its intended role is to prioritise cases and surface a candidate recommendation for a team that retains the decision.

1.3 Aim and Objectives

The aim is to design, implement and evaluate an end-to-end machine-learning pipeline that predicts breast cancer MDT treatment recommendations and remains applicable across institutions with differing data conventions. This aim is pursued through four objectives, each answered by a component of the evaluation in Chapter 5.

1. **Feature-set design.** Construct three nested feature sets (base, intermediate and full) reflecting varying levels of data availability, and evaluate how predictive performance scales with data richness.
2. **Data harmonisation.** Develop a configurable harmonisation layer mapping differing column names and value encodings onto a common schema, enabling a model trained at one institution to be applied at another without redesign.
3. **Cross-institutional and cross-continental evaluation.** Evaluate generalisation from KATH to SGMC and from KATH to SEER, using recorded survival outcomes to examine whether the transferred model produces clinically coherent behaviour where MDT labels are unavailable.
4. **Reproducible pipeline delivery.** Deliver a documented, reproducible end-to-end pipeline; and, as an extension, design and implement a retrieval-augmented generation (RAG) layer that grounds recommendations in clinical guideline text, evaluated as a proof-of-concept.

These objectives correspond to four research questions.

RQ1: can a classifier predict the recorded MDT recommendation at useful accuracy, and how does that depend on the number and completeness of available variables?

RQ2: does the granularity of the target materially change achievable performance, and is a coarser, clinically grounded target more defensible?

RQ3: can a configurable harmonisation layer carry a model to a second hospital whose schema and encodings differ, without retraining?

RQ4: does the model retain coherent behaviour when applied to a different continent, and can survival outcomes serve as an external proxy for clinical validity where MDT labels are absent? The scope is bounded to breast cancer, to retrospective evaluation with no clinician involved in training or in live decisions, and to the RAG layer as a proof-of-concept.

1.4 Contributions

- One of the first applications of ML-based MDT recommendation prediction to real-world breast cancer data from an African healthcare setting.
- A configurable, extensible harmonisation layer reconciling cross-institutional schema and value differences, demonstrated across three independently sourced datasets.
- A cross-continental transfer evaluation that uses recorded survival outcomes to test the behavioural coherence of a transferred model, together with a measured account of how far the receiving population differs from the source.
- A controlled re-analysis establishing that predictive performance in this setting is governed principally by data richness rather than classifier choice, and a demonstrated method for exposing an unlearnable target class and an indiscriminate evaluation instrument.

1.5 Significance

The findings locate the binding constraint on decision-support performance in this setting. If performance is governed by the richness and completeness of recorded data rather than by the choice of classifier, then institutions considering decision support should invest in data capture before modelling capability, which is directly actionable for the lower-resource hospitals the work targets. The harmonisation layer occupies a space the reviewed literature leaves empty: schema heterogeneity is treated either as out of scope, as in single-institution MDT studies, or as a problem requiring enterprise-scale standardisation, as in the common data model tradition. Demonstrating that a lightweight, inspectable mapping is sufficient to carry a model across institutions retains the interoperability principle at a fraction of the cost, and provides the prerequisite infrastructure that local validation at a receiving hospital requires. Finally, the candid

treatment of a withdrawn agreement figure and a confounded survival association demonstrates an evaluation discipline that is itself a contribution to how such feasibility studies should be reported.

1.6 Report Structure

Chapter 2 reviews related work thematically and identifies the collective gap this project addresses. Chapter 3 presents the methodology: research design, harmonisation strategy, target definition, modelling and evaluation, each justified against the literature. Chapter 4 combines system design, implementation and experimental design, showing how the research gap drives each design choice. Chapter 5 reports and critically evaluates the results across the three tasks, the imbalance and calibration analyses, the external transfers and the RAG extension. Chapter 6 concludes and identifies future work.

2 Literature Review

This review builds an argument rather than cataloguing papers. Section 2.1 gives the domain background; Section 2.2 sets out the technical approach adopted; Sections 2.3 to 2.7 synthesise the existing literature around themes of method, data, evaluation and transfer; and Section 2.8 draws these together into the collective research gap that motivates the dissertation.

2.1 Background

MDT meetings constitute the standard of care in breast cancer management [1], and the consistency and quality of their decisions bear directly on patient outcomes. The process is resource-intensive and contingent on the availability of experienced specialists, a dependency that becomes a structural constraint in under-resourced systems. These pressures have driven research into whether artificial intelligence can augment MDT decision-making; the systematic review and meta-analysis of AI decision-support systems for tumour diseases reports wide variation in accuracy across systems and tumour types, and attributes much of that variance to heterogeneity in evaluation practice rather than genuine differences in capability [7]. Two lines of work are relevant: classical ML classifiers applied to structured clinical data, and Large Language Models (LLMs) applied to oncology decision support. A consistent limitation, evaluation confined to single institutions and small samples, recurs across both and motivates the cross-institutional focus of this dissertation.

A second strand of background concerns the reconciliation of heterogeneous data. Hospitals encode equivalent clinical concepts under different field names, value sets and levels of granularity, so a model trained on one schema cannot be applied to another without translation. Health informatics has long addressed this through the common data model: the Observational Medical Outcomes Partnership (OMOP) model maps heterogeneous source databases onto a shared relational schema and standardised vocabularies, validated across ten independent observational databases [12], and the Observational Health Data Sciences and Informatics (OHDSI) collaboration scaled this to hundreds of millions of records [11]. These establish translation to a shared representation as the accepted route to interoperable multi-institutional research, a principle this project retains while scaling the machinery to a bounded task.

2.2 The Technique: Tree Ensembles for Structured Clinical Prediction

The task is a multi-class classification problem over structured, predominantly categorical clinical variables, in which the treatment decision depends on interactions between variables rather than on any single one. Two properties of the data therefore determine the

appropriate classifier family. Tree ensembles partition on categorical values directly, represent interactions natively, and require neither a distance metric nor an assumption of feature independence, which is precisely where distance-based, kernel-based and Naive Bayes methods are weakest on this kind of data.

A review of 48 disease-prediction studies found no single algorithm dominant across tasks: the Support Vector Machine was most frequently applied, followed by Naive Bayes, with Random Forest among the most accurate where used [34]. Each family has conditions under which it performs well and conditions under which it does not. Decision trees are transparent and handle categorical variables and non-linear interactions without preprocessing, but a single tree overfits and is unstable on modest cohorts. Support Vector Machines perform well where classes are separable in a well-chosen feature space, but require continuous, scaled inputs and extend to multi-class problems only indirectly. K-Nearest Neighbours depends on a meaningful distance between cases and degrades in the high-dimensional sparse spaces produced by one-hot encoding. Naive Bayes is fast with limited data, but its conditional-independence assumption is violated when stage, receptor status and grade are correlated, as they typically are. Neural networks can achieve high accuracy, and a breast-cancer comparison reported a neural network as strongest on a well-structured diagnostic dataset [35], but they require larger training volumes and offer limited interpretability.

Within the tree-ensemble family, Random Forests aggregate many decorrelated trees through bootstrap aggregation, conferring robustness to noise and an intrinsic variable-importance measure [15]; gradient-boosted trees, as realised in XGBoost, add regularised stage-wise optimisation and native handling of missing values [16], advantageous where clinical records are incompletely populated. These properties account for the recurrent preference for ensemble classifiers in structured MDT prediction and inform the modelling choices in Chapter 3. The alternatives are set aside not because they are weak in general, but because their strengths do not align with the structure of this data.

The explanation extension adopts a distinct technique, retrieval-augmented generation (RAG), in which a retriever selects passages from a trusted document in response to a query and a generation model is constrained to answer using only those passages [38]. This directly operationalises the recurring recommendation in the medical-LLM literature that model outputs be grounded in authoritative external text rather than trusted in isolation [18]. The technique is applied here for a narrow purpose: the classifier remains the sole decision-maker, and the RAG layer is asked only to explain a prediction with reference to a guideline, never to propose or override a recommendation.

2.3 Classical and LLM-Based MDT Prediction

The most directly comparable prior work is the CASCADE system [2], which applied an XGBoost classifier to predict MDT recommendations for hepatocellular carcinoma from 140 patients across eight categories, reporting predictiveness across all eight and greater accuracy than recommendations derived from published ESMO and NCCN guidelines. CASCADE establishes that tree-based ensembles can reproduce complex multi-category MDT decisions at a clinically credible level and validates the framing adopted here, but its limitations define the space this dissertation occupies: single-institution training and evaluation, no mechanism for reconciling schema differences, no external test, and a focus on liver rather than breast cancer. A closely related study by Sondermann et al. predicts prostate-cancer tumour-board recommendations from 1,929 records using Decision Tree, Random Forest and K-Nearest Neighbours classifiers [3]; Random Forest achieved the highest accuracy at 66.3%, with near-perfect prediction of frequent recommendations but markedly weaker performance on rarer categories, attributed to class imbalance. This pattern, strong on common classes and degrading on rare ones, anticipates a central challenge of the present work.

General-purpose LLMs applied to breast cancer MDT settings have produced consistently qualified results. Sorin et al. tested GPT-3.5 on ten cases from a single institution and reported concordance with the actual MDT decision in roughly 70% of cases, constrained by the small sample and the absence of guideline grounding [4]. Griewing et al. reached a more sceptical conclusion, reporting 50.0% overall concordance and 58.8% for breast cancer across twenty constructed patient profiles, with several recommendations judged clinically inappropriate [5]. Benary et al. evaluated LLMs for treatment-option generation using ten fictional molecular tumour-board cases, finding plausible options that often fell outside current guidelines and were poorly supported by verifiable evidence [6]. Read together, these studies bound the achievable performance of ungrounded LLMs at roughly 50 to 70% concordance on small, same-institution or constructed samples, and share a decisive weakness: limited case numbers, no cross-institutional testing, and no integration with structured ML classification. These MDT-specific findings sit within a broader evidence base: general-purpose models adapted for medicine, notably Med-PaLM, encode substantial clinical knowledge and approach expert performance on question-answering benchmarks [17], but surveys consistently identify a tendency to produce fluent yet unverifiable statements and unresolved questions of safety and accountability [18], which is what motivates the grounded RAG extension.

A distinct body of work applies LLMs not to recommendation but to extraction, pulling structured variables out of free-text notes. A scoping

review of this literature in oncology identifies 24 eligible studies, finds most evaluate BERT-family models fine-tuned on clinical text, reports breast cancer among the most frequently studied types, and finds extraction strong on well-defined variables but degrading on those requiring clinical inference, with very few studies validating beyond a single institution [37]. This literature matters here by contrast: the datasets used are already structured, so no extraction is required, and its single-institution limitation corroborates the same gap identified in the structured-prediction literature. Table 2.1 summarises the directly comparable studies.

Table 2.1: Reviewed MDT prediction studies. All prior studies evaluate internally only; none is externally validated. Figures are as reported in the original publications.

Study	Institution(s)	Sample	Method	Reported accuracy	Ext. val.
CASCADE [2]	Single	140	XGBoost	Predictive across 8 categories	No
Sondermann et al. [3]	Single	1,929	DT / RF / KNN	66.3% (RF)	No
Sorin et al. [4]	Single	10	GPT-3.5 (ungrounded)	~70% concordance	No
Griewing et al. [5]	Constructed	20	GPT-3.5 (ungrounded)	50.0% / 58.8% breast	No
Benary et al. [6]	Fictional	10	LLMs (ungrounded)	Often outside guidelines	No
This dissertation	3 institutions, 2 continents	5,000 / 1,000 / 4,024	RF / HistGB + harmonisation	86.73% (6-group)	Yes (2)

2.4 Data Harmonisation Approaches

Common data models are not the only response to schema heterogeneity, and reviewing the alternatives clarifies why a lightweight approach is defensible here. A schema-level tradition maps heterogeneous cohorts onto a standard schema: Almeida et al. present a semi-automatic methodology inspired by OHDSI but scaled to disease-specific study, achieving complete or close correspondence for 111 of 120 variables (roughly 93%) when harmonising Alzheimer's cohorts across countries, with the remainder harmonised at a marginal loss of granularity [31]. This is the same

trade-off the value mappings in this project accept, and it establishes schema-level harmonisation with documented, partial information loss as accepted practice rather than a compromise peculiar to this work. More recent work automates the matching: Yang et al. learn semantic

representations from variable descriptions and distributional representations from participant data, scoring variable pairs by similarity so equivalent fields can be aligned without a manual dictionary [32]. Automation is attractive at scale, but requires informative descriptions and sufficient overlapping data, and its matches are probabilistic rather than clinically verified; for a few dozen breast-cancer variables whose correspondences a clinical collaborator can confirm directly, a transparent dictionary is both sufficient and more auditable. A third tradition, prominent in imaging and genomics, treats harmonisation as the statistical removal of non-biological variation: ComBat corrects batch effects using an empirical Bayes framework [33], but is not applicable here, whose heterogeneity is representational (differing names and encodings) rather than a distributional shift in a shared numeric feature. Naming this tradition delimits the kind of harmonisation this project performs: reconciliation of schema and vocabulary, not statistical correction of measurement bias. The proportionality argument follows: whereas OMOP pursues exhaustive standardisation, the narrow variable set and modest scale of the MDT task make a lightweight, configurable mapping the more appropriate choice.

2.5 Feature Selection and Data Availability

A question that runs through the CASCADE and prostate-cancer studies but is rarely posed explicitly is how much data a predictive model actually needs. Both fix a variable set determined by what their institution happened to record and report performance for that set alone [2,3]; neither examines how performance would degrade at a hospital recording less. For a system intended to travel between institutions with unequal data-capture maturity, this is the central practical question. The foundational treatment of feature selection distinguishes its goals, improving prediction, reducing cost, and providing a more interpretable account, and makes two arguments that bear on the present design [13]: reducing dimensionality can improve generalisation because irrelevant variables contribute variance without signal, and the usefulness of a variable is defined only relative to the others present, so subsets must be evaluated empirically rather than assembled by ranking variables one at a time. Surveys organise the techniques into filter methods (intrinsic statistical criteria, model-agnostic but blind to interactions), wrapper methods (search by classifier performance, capturing interactions but expensive and prone to overfitting the selection), and embedded methods (selection during training) [14].

A review of feature selection for disease-risk prediction observes that clinical datasets typically present far more candidate variables than patient records, the high-dimension, low-sample-size regime in which each method risks overfitting the selection to the development cohort, and notes that clinically motivated selection tends to produce models

that practitioners trust and adopt more readily [36]. The design adopted here is best understood as a constrained wrapper: rather than searching all subsets, which would optimise for the training institution and produce a variable set no other hospital could supply, three nested sets are fixed in advance by what hospitals at different levels of data-capture maturity can realistically record, and each is evaluated end-to-end. This sacrifices any claim to an optimal subset in exchange for a directly interpretable answer to the deployment question, and treats feature selection as an experimental variable rather than a preprocessing step.

2.6 Evaluation Under Class Imbalance

The reviewed studies report performance in ways that are not straightforwardly comparable, and the meta-analysis attributes much of the variance it observes to this heterogeneity of evaluation practice rather than to genuine differences in capability [7]. The recurring hazard is class imbalance. MDT recommendation distributions are inherently skewed: a few standard pathways account for most patients, while clinically important but uncommon decisions appear rarely. Overall accuracy on such a distribution is dominated by the frequent classes, so a model that predicts the majority pathway can post a respectable accuracy while being useless for exactly the cases where decision support has most value; the prostate-cancer study documents this directly [3]. Two responses follow, both adopted here. The first is to report metrics that weight classes equally: macro-averaged F1 averages per-class F1 without weighting, so a class the model never recovers drives the score down regardless of size, and balanced accuracy averages per-class recall to the same end. The second is stratified partitioning, so rare categories appear in each split in proportion to their prevalence. Where imbalance is severe enough that metric choice alone is insufficient, algorithmic remedies apply: SMOTE synthesises minority instances by interpolating between existing ones rather than duplicating them [21], and broader surveys organise resampling alongside cost-sensitive learning and ensemble strategies, observing that no single technique dominates [22]. These remedies are evaluated directly in this project rather than left as future work.

2.7 External Validation, Transportability and Dataset Shift

The claim that a model works is ambiguous until the validation design is specified. The clinical prediction literature distinguishes internal validation, which estimates performance on unseen patients from the same population, from external validation, which estimates performance on a different population, institution or period and determines whether a model generalises beyond the setting that produced it [19]. The influential seven-step development framework treats validation as a structured assessment of calibration and discrimination in new settings and is explicit that internal validation

cannot substitute for the external case [24]; a parallel account frames the sequence as external validation, then updating, then impact assessment, on the reasoning that a model failing externally should be adapted rather than abandoned [25]. The empirical picture is discouraging and is the strongest justification for this design: a systematic examination of newly published risk-prediction models found external validation performed infrequently, rarely by independent researchers, and typically revealing worse discrimination than the development cohort [20]. The pattern is directly visible in the MDT literature: CASCADE and the prostate-cancer study report internal estimates only [2,3], and the LLM evaluations report concordance on small same-institution or constructed samples [4,5,6]. None establishes what would happen at a second hospital, which is precisely the question a system intended for lower-resource settings must answer.

Interpreting an external result requires knowing how far the validation setting differs from the development setting, since a performance drop is uninformative unless the relatedness of the two populations is characterised [26]. The two external evaluations here are deliberately of different kinds: the transfer to SGMC holds the healthcare system roughly constant while varying schema and recording conventions, whereas the transfer to SEER varies population, coding standard and available variables simultaneously. These are two points on a spectrum of relatedness, and reporting them separately is what allows the schema problem to be distinguished from the population problem. The mechanism behind external performance loss is dataset shift, the condition in which the deployment distribution differs from the training one; a widely cited clinical account distinguishes shift arising from changes in patient population, in clinical practice, and in data-capture technology, and argues such shifts are a principal driver of AI malfunction in medicine [27], while formal treatments separate shift in the predictors from shift in the predictor-outcome relationship [28]. This taxonomy sharpens what the project tests: harmonisation addresses only the representational component of shift, the fact that the same concept is written differently; it does nothing about the substantive components, namely that SGMC's and SEER's patients genuinely differ from KATH's. A model can therefore be perfectly harmonised and still transfer poorly, and separating these failure modes is essential to interpreting Chapter 5 correctly.

A more sceptical position questions the goal itself, arguing that generalisability as commonly conceived is a myth because site heterogeneity is a real property of medical practice rather than noise, and that the productive response is to build infrastructure allowing models to be validated and tailored locally rather than to pursue one universal model [29]. This deserves to be met rather than ignored. The position taken here is that the two views are reconcilable and that the

harmonisation layer is better understood in the sceptics' terms: before a receiving hospital can evaluate, retrain or tailor a model on its own data, its data must be expressible in the model's vocabulary at all, and that translation is exactly the prerequisite the local-tailoring position presupposes but does not itself provide. The contribution claimed is therefore not a model that generalises everywhere, but a mechanism that makes local evaluation possible without redesign.

2.8 Identified Gaps and Research Positioning

Considered collectively rather than as isolated limitations of individual papers, the reviewed studies achieve three things: they establish that tree-based classifiers can match or exceed clinical guidelines for MDT prediction on real but single-source data [2,3], that ungrounded LLMs remain limited to 50 to 70% concordance on small samples [4,5,6], and that mature templates exist for reconciling schema heterogeneity in observational research [11,12]. What they consistently fail to address defines the gap. Table 2.2 sets out the correspondence between what the literature leaves unresolved and what this dissertation does in response.

Table 2.2: Collective limitations of the reviewed literature and the response adopted in this dissertation.

Collective limitation	Evidence across studies	Response in this work
No African real-world MDT data	Work concentrated in high-income institutions [2,3,7]	Real MDT records from KATH and SGMC (Ghana)
Schema heterogeneity unaddressed for MDT	Common data models mature [11,12] but never applied to MDT prediction	Lightweight configurable harmonisation layer
No external validation	All reviewed MDT studies internal only [2,3,4,5,6]; external validation rare and worse where done [20]	Two external transfers, KATH to SGMC and to SEER
Dataset shift not confronted	No reviewed MDT study characterises transfer [27,28]	Membership AUC and per-variable PSI quantify relatedness and residual shift
Data-need question unposed	Fixed variable sets, no degradation analysis [2,3]	Nested feature sets as an experimental variable
Ungrounded LLM outputs	Fluent but unverifiable recommendations [4,5,6,18]	RAG layer grounding explanations in a licensed guideline

These gaps matter because the institutions that stand to benefit from decision support are, by definition, not the institutions that produced the training data, so a system that has never been tested outside its source cannot be claimed to serve them. The transition from existing research to the present work is therefore direct: existing studies establish feasibility within a single institution but leave cross-

institutional transfer, schema reconciliation and external validation collectively unaddressed; this dissertation navigates those limitations by treating harmonisation as a first-class concern, by fixing feature sets so the data-need question can be answered, and by reporting two external transfers with their relatedness measured rather than assumed. The literature review therefore establishes a clear justification for the methodology and experimental design that follow.

3 Methodology

This chapter sets out the design decisions and their justification against the literature; Chapter 4 documents their implementation and experimental setup. Wherever a diagram, flow or table communicates a decision more efficiently than prose, it is used in place of extended text, with only the explanatory detail needed to interpret it retained.

3.1 Research Design and Process Model

The project follows the Cross-Industry Standard Process for Data Mining (CRISP-DM), the de-facto standard for applied data-mining work [10]. It is preferred over alternatives such as SEMMA, which omit an explicit business-understanding phase, an omission inappropriate here since the pipeline must reflect the specific data constraints of the healthcare provider. A waterfall rather than agile application was adopted because access to the clinical domain expert is limited, making the rapid per-phase feedback an agile approach assumes impractical. Figure 1 maps the six phases onto the dissertation.

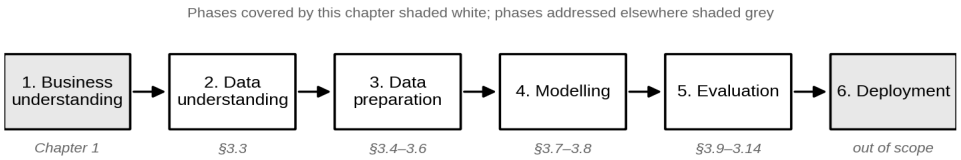


Figure 1: The six phases of CRISP-DM as applied. This chapter covers phases 2 to 5; phase 1 is addressed in Chapter 1, and phase 6 (deployment) is out of scope.

The study is a quantitative, retrospective, experimental evaluation on historical clinical records. It is retrospective because the decisions being predicted were made and recorded before the project began, which permits an unbiased comparison of prediction against recorded outcome but precludes any claim about influence on care. It is experimental in that the pipeline is executed repeatedly under systematically varied conditions, feature richness, target granularity and receiving institution, with all other elements held as constant as the design allows. Two ordering constraints carry methodological weight: harmonisation precedes any modelling, because fitting first and reconciling schemas afterwards is what confines existing studies to a single institution [2,3]; and partitioning precedes fitting, with the test partition untouched until final evaluation, so no information from held-out patients can influence the model.

3.2 Data Understanding

The project draws on three datasets. KATH (Komfo Anokye Teaching Hospital, Ghana) contains 5,000 breast cancer MDT records described by 80 variables, with a single combined outcome variable recording the MDT's final recommendation. SGMC (Sweden Ghana Medical

Centre, a Ghanaian referral centre collaborating with KATH) contains 1,000 records described by 121 variables, encoding the MDT decision across several separate columns rather than as a single label. SEER (United States) comprises 4,024 records described by 21 variables; it contains no explicit MDT recommendation and omits HER2 status, but includes real survival outcomes. Initial exploration highlighted two properties that shape preprocessing: the outcome categories are unevenly distributed, and the three datasets record equivalent clinical concepts using different column names and value conventions, the interoperability barrier common data models were devised to overcome [11,12].

The choice of these three sources follows from the objectives. A single-institution design using only KATH was rejected because it cannot test cross-institutional transfer at all. A search for other public MDT-recommendation datasets to supplement or replace SGMC or SEER found none, itself consistent with the gap this project identifies, since institutions publishing MDT prediction work rarely release the underlying data. SEER was included because it is the largest publicly documented alternative and because its US provenance and absence of an MDT label provide a maximally demanding test of transfer, to a different continent, coding standard and outcome representation, which a second Ghanaian comparison could not provide. The three sources are therefore the most demanding combination available for the cross-institutional claim, not an arbitrary convenience sample.

3.3 Data Preparation

Preprocessing loads each dataset, inspects its structure and value formats, removes rows with missing values in the variables of interest by listwise deletion, and converts categorical variables to a numerical representation by one-hot encoding. Each step reflects a deliberate trade-off. Listwise deletion was chosen for the initial complete-case dataset because, under a missing-completely-at-random assumption, it yields an unbiased sample and avoids imputation artefacts before a baseline is established; the cost is reduced sample size, and the alternative of early imputation was deferred so that the influence of data completeness could be examined directly through the nested feature sets. One-hot encoding was preferred to ordinal integer encoding because integer coding would impose a spurious ordering on unordered categories such as tumour subtype, which split- and distance-based learners can misinterpret as magnitude. Complete-case filtering is applied after feature selection rather than before, which preserves the maximum cohort for each feature set; the consequence, that configurations are evaluated on different cohorts, is the confound the controlled design in Section 3.6 removes.

3.4 Harmonisation Strategy

Harmonisation is required at two distinct levels, and conflating them is a common source of silent error. At the schema level, the same concept carries different field names: KATH's Age is SGMC's age_years and SEER's Age; KATH's AJCC_Stage_Group is SGMC's stage_group_derived and SEER's 6th Stage. At the value level, even once two fields are known to denote the same concept, they may encode it differently: HER2 status is a clinical category at KATH (Positive, Negative, Equivocal, Pending) but a raw immunohistochemistry score at SGMC (0, 1+, 2+, 3+); stage is Stage II at KATH, II at SGMC and the substage IIA at SEER. Renaming alone would therefore produce a dataset that is structurally aligned and semantically incoherent, and the failure would be silent: a model would encode 3+ and Positive as two unrelated categories and learn nothing from either. The layer is consequently implemented as two ordered stages, column mapping then value mapping, shown in Figure 2. Table 3.1 gives representative examples; the full mapping is in Appendix A.

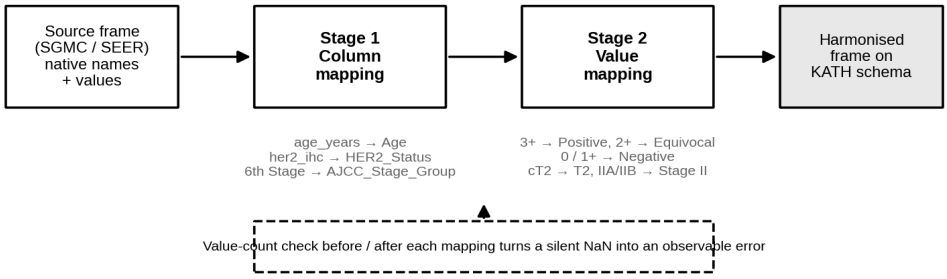


Figure 2: The two-stage harmonisation flow. A value-count check before and after each mapping converts a silent failure (an unmapped value becoming NaN) into an observable one.

Table 3.1: Representative column-name and value-vocabulary differences the harmonisation layer reconciles.

Clinical concept	KATH	SGMC	SEER
Age (column)	Age	age_years	Age
Stage group (column)	AJCC_Stage_Group	stage_group_derived	6th Stage
Stage (value)	Stage II	II	IIA (substage)
HER2 (encoding)	Positive / Negative / Equivocal / Pending	0 / 1+ / 2+ / 3+ (IHC)	Not recorded

Clinical concept	KATH	SGMC	SEER
T category (value)	T2	cT2	T2 (varies)

The value mappings encode recognised clinical conventions rather than arbitrary format conversions. They were derived by the author from published sources. The HER2 mapping (0 and 1+ to Negative, 2+ to Equivocal, 3+ to Positive) reflects standard clinical practice [51], in which a 2+ result is regarded as equivocal pending reflex in-situ hybridisation and only a 3+ score is a definitive positive. The stage mapping collapses SEER's substages onto KATH's main groups, so IIA and IIB both become Stage II. This is a lossy transformation, acknowledged as such: substage encodes prognostically meaningful information about tumour and nodal burden, and discarding it is a cost accepted in exchange for a shared vocabulary. The reverse is not available, because KATH never recorded substage; harmonisation can only level down to the coarsest source. Section 5.6 returns to this as a clinically identified limitation.

Implementing this as a lightweight, configurable dictionary rather than adopting OMOP is justified by proportionality. The task requires a few dozen breast-specific variables, so most of a common data model's scope would be unused; OMOP entails mapping source values onto standardised vocabularies, an effort measured in months and reported as the principal difficulty even in its own validation study [12]; and the mapping tables must be legible to, and correctable by, the clinical collaborator, which a plain dictionary supports and a vocabulary pipeline does not. The principle drawn from OHDSI, that translation to a shared representation is what makes cross-institutional modelling possible [11], is retained; only the machinery is scaled to the problem.

3.5 Feature-Set Design and Target Definition

The three nested feature sets operationalise feature selection as a first-class experimental variable. They are defined by the level of data a hospital can realistically supply rather than by an automated search over subsets, so the resulting comparison measures how predictive performance scales with data richness while keeping every configuration clinically deployable. Being nested is a load-bearing property exploited by the controlled design in Section 3.6: every patient complete on all thirty-two variables is necessarily complete on the fifteen and the five.

The target is reported in two forms rather than choosing between them. KATH records the MDT decision in a single field with fifteen categories that is severely imbalanced (neoadjuvant chemotherapy accounts for several hundred test cases while adjuvant trastuzumab-based therapy accounts for four) and partly redundant, since several categories

describe the same management strategy at different points in the pathway. The strict fifteen-class task predicts the exact category and is retained as the harder baseline. A grouped six-class task collapses the fifteen categories into clinically coherent super-groups (Systemic-Chemo, Surgery, Radiotherapy, Endocrine, Palliative/Supportive and Further-workup) on the basis of the treatment modality decided upon, which is closer to the triage question of which pathway a patient enters. Reporting both is a matter of honesty: collapsing classes necessarily raises accuracy because confusions internal to a group become correct answers, and a grouped figure presented alone would invite comparison against fifteen-class results elsewhere and overstate the contribution. One consequence is anticipated and tested in Chapter 5: Further-workup is a residual category collecting repeat biopsy, further imaging, genetic counselling and clinical trial consideration, and is therefore unlikely to possess a coherent clinical signature. Table 3.2 summarises both.

Table 3.2: Nested feature sets and the two target definitions.

Element	Size	Definition
Base feature set	5 variables	Age, stage, ER, PR, HER2 status
Intermediate set	15 variables	Base plus ten further recorded variables
Full set	32 variables	Intermediate plus mammogram BIRADS, Ki67 and further fields
Fifteen-class target	15 classes	Exact recorded MDT_Final_Recommendation; severely imbalanced
Six-group target	6 groups	Systemic-Chemo, Surgery, Radiotherapy, Endocrine, Palliative/Supportive, Further-workup

3.6 Model Development and Controlled Comparison Design

Two classifier families are used, a Random Forest and a histogram-based gradient boosting classifier, both tree ensembles for the reasons given in Section 2.2. The Random Forest serves as the baseline, being the model reported strongest in the most closely comparable study [3] and supplying a variable-importance measure and robustness through decorrelation [15]; gradient boosting fits trees stage-wise to residual error with explicit regularisation, typically achieving higher accuracy on richer structured data at the cost of greater hyperparameter sensitivity [16].

The original analysis assigned a Random Forest to the base set and gradient boosting to the richer sets, which means model type and feature count varied together and no difference could be attributed to feature richness alone. A second confound is more serious: because complete-case filtering follows feature selection, each configuration is evaluated on a different cohort, and if fully documented cases were also systematically easier to predict, an apparent gain from richer data

would in fact measure cohort selection. A controlled re-analysis removes both, following the requirement that a comparison hold the design fixed and vary one factor at a time [24]. Because the feature sets are nested, a common cohort of records complete on the full set exists without further filtering; the partition is computed once on that cohort and applied to each configuration by index, guaranteeing identical patients in every run, and both classifiers are fitted across all three feature sets. Two costs are accepted: the controlled analysis is silent about records excluded by complete-case filtering, and it uses less data than the original. Both original and controlled analyses are reported, since the difference between a confounded and a controlled estimate of the same quantity is itself informative and the reporting standards in Section 3.8 require the analysis actually conducted to be visible [23].

3.7 Evaluation Strategy

The evaluation is structured as three tasks of increasing difficulty, each removing a further assumption. Task 1 is an internal evaluation on KATH, estimating performance on unseen patients from the same distribution (RQ1, RQ2). Task 2 is an external cross-hospital evaluation applying the KATH-trained model to SGMC without retraining, where distribution, schema and outcome recording all change (RQ3). Task 3 is an external cross-continental evaluation on SEER, where population, coding standard and available variables all differ and no MDT label exists (RQ4).

For Task 1, KATH is partitioned 70/15/15 into training, validation and test sets, stratified by the target so rare categories appear in each partition in proportion to their prevalence. Stratification is essential given the imbalance; a safeguard falls back to an unstratified split when any class has fewer than two members, which can occur on the full feature set once filtering has reduced the cohort. Performance is reported as accuracy, macro-averaged F1 and balanced accuracy together, because accuracy alone is dominated by frequent categories and would conceal failure on rare ones. A confusion matrix and per-class report are produced for the strongest configuration, because the clinical seriousness of an error depends on what it is confused with: confusing surgery with chemotherapy in a borderline operable patient is a different kind of error from confusing palliative care with curative surgery. Five-fold stratified cross-validation is applied to the strongest configuration to establish that the reported figure reflects the model rather than a fortunate partition.

Tasks 2 and 3 cannot use accuracy, because neither receiving institution records the target in KATH's form. SGMC records its decision across several fields (chemotherapy, surgical plan, radiotherapy, endocrine therapy, intent), so a rule-based agreement check determines whether the model's single prediction is consistent

with the combination SGMC recorded, scored against trivial constant predictors as a designed base-rate baseline. SEER contains no MDT recommendation but does contain survival months and vital status, so predictions are examined against survival with a chi-square test of association between predicted group and vital status. The evidential weight of this last procedure is limited and structural: an association between predicted management and survival is expected even from a model that merely recovers disease stage, because stage drives both, so the survival analysis is read throughout as a check on behavioural coherence rather than as evidence of clinical correctness.

3.8 Robustness, Validation and Reporting

This section consolidates the class-imbalance treatment, calibration assessment, external-validation interpretation and reporting standards into a single account of the checks the study applies; Table 3.3 summarises the configuration, and the implementation of each is in Chapter 4.

Class imbalance. Two remedies representing the two dominant families are evaluated: cost-sensitive learning through inverse-frequency class weighting, which changes only the training objective and invents no data, and SMOTE, which synthesises minority instances by interpolation [21,22]. Both are applied on the controlled cohort and partition, and to the training split only, since synthesising or reweighting test instances would measure performance on a distribution that does not exist. The expected outcome is stated in advance, that rebalancing recovers rare classes at the expense of frequent ones so accuracy falls while macro-F1 rises, because stating the expectation determines how a departure from it should be read; Section 5.4 reports that the expected trade did not occur. SMOTE's neighbour parameter is bounded by the smallest training class, a binding rather than theoretical constraint given the imbalance.

Calibration. Every discrimination metric measures whether the model assigns the correct label; validation also requires calibration, whether reported probabilities match observed frequencies [24], because the system is intended to surface a confidence alongside each recommendation and a model confident and wrong more often than its confidence implies is actively misleading [17,18]. Two measures are adopted: the multi-class Brier score, a proper scoring rule penalising poor discrimination and poor calibration together, and expected calibration error, which bins predictions by top-class confidence and averages the gap between confidence and accuracy per bin. A reliability diagram is reported alongside both, since the aggregate can conceal whether a gap comes from a few difficult cases or from ordinary behaviour.

External-validation interpretation and dataset shift. A performance drop is uninformative unless the relatedness of the two populations is characterised [26]. A membership model, a classifier trained to distinguish development from validation patients on shared covariates, converts each transfer into a measured point on a relatedness scale: an AUC near 0.5 means a drop implicates the model, while an AUC near 1.0 means the populations differ enough that a drop is expected regardless of model quality. Residual shift after harmonisation is measured per variable using the Population Stability Index (below 0.10 negligible, 0.10 to 0.25 moderate, above 0.25 major), which identifies where the model is extrapolating [27,28]. This makes explicit that harmonisation addresses only the representational component of shift, consistent with the sceptical position that the layer is best offered as the prerequisite for local evaluation rather than as a mechanism for transferring a model unchanged [29].

Reporting standards. Reporting follows the TRIPOD checklist for retrospective prediction-model studies [23,30]: participant flow with reasons for exclusion is reported per feature set, development and external performance are reported separately, the missing-data method and its rationale are stated, predictors are restricted to variables recorded before the MDT meeting, and where an analysis was superseded both are reported, since selective presentation is among the practices that make such studies difficult to trust [30].

Table 3.3: Configuration of the robustness, calibration and shift checks.

Check	Method	Purpose / threshold
Imbalance	Inverse-frequency weighting; SMOTE	Recover rare classes; training split only
Discrimination	Accuracy, macro-F1, balanced accuracy	Expose failure on rare classes
Calibration	Multi-class Brier; expected calibration error; reliability diagram	Confidence must match observed accuracy
Relatedness	Membership-model cross-validated AUC	0.5 comparable; 1.0 extrapolation
Residual shift	Population Stability Index per variable	<0.10 negligible; >0.25 major
Reporting	TRIPOD checklist	Separate internal / external; report superseded analyses

3.9 Design of the Retrieval-Augmented Explanation Layer

The medical-LLM literature identifies fluent but unverifiable recommendations as the recurring weakness of ungrounded models, with the recurring remedy being to ground outputs in authoritative external text [18]. RAG operationalises this: a retriever selects

passages from a trusted document and a generation model is constrained to answer using only those passages [38]. The classifier remains the sole decision-maker; the RAG layer is given the classifier's prediction and asked only to explain it with reference to a guideline, never to propose or override a recommendation, which keeps the validated classifier in control and confines the language model to the narrower, more checkable task of producing a citation-grounded rationale.

The grounding guideline is NICE NG101, Early and Locally Advanced Breast Cancer: Diagnosis and Management [44], chosen because its open licence permits use with an AI model (a necessary condition, since the text is indexed and retrieved directly) and because it is written as numbered recommendations rather than algorithm diagrams, which extract as more reliably coherent text than a flowchart layout. Evaluating a generated explanation requires two questions to be kept separate: whether the classifier's recommendation agrees with what the guideline advises (concordance), and whether the explanation confines itself to what the retrieved passages actually say (faithfulness). An automated faithfulness check comparing explanation words against retrieved-passage words proved unreliable, flagging ordinary paraphrase (writing therapies where the source says treatment) as often as genuine fabrication. Rather than report a number known to be unreliable, every explanation was read and classified by hand, and Section 5.7 reports that manual review. This reflects the reporting standard applied throughout: where an automated measure was found not to answer the question asked, it was replaced rather than reported uncritically.

Chapter Summary. This chapter set out the methodology's design decisions and their justification. The pipeline follows CRISP-DM as a retrospective, quantitative, experimental study, with a two-stage harmonisation layer reconciling column names and value encodings across KATH, SGMC and SEER. Three nested feature sets (five, fifteen and thirty-two variables) and two target granularities (fifteen-class and six-group) were defined as experimental variables rather than fixed choices, and a controlled comparison design was introduced to separate the effect of feature richness from classifier choice. The evaluation strategy spans three tasks of increasing difficulty, supported by robustness checks for imbalance, calibration, and external-validation interpretation.

4 System Design, Implementation and Experimental Setup

This chapter documents how the design of Chapter 3 was built and how the experiments were constructed. It is organised so that the progression from research gap to design choice to implementation to experimental setup to evaluation strategy, shown in Figure 4, is explicit: each design choice in the pipeline traces to a gap identified in Chapter 2. Four sections follow: the pipeline and its environment (4.1), the harmonisation and preprocessing implementation (4.2), the experimental setup and evaluation components (4.3), and the RAG implementation (4.4). An implementation checklist (Table 4.1) records that all required programming elements were considered.

4.1 Pipeline Architecture and Environment

The pipeline is implemented in Python 3.12 in a Google Colab notebook, with the three datasets held in Google Drive and mounted at runtime. The scientific stack is pandas [45] and NumPy [46] for data manipulation, scikit-learn [47] for modelling and evaluation, SciPy [48] for the statistical test and Matplotlib [49] for figures; openpyxl reads the two Excel-formatted hospital extracts. A hosted notebook was chosen because the datasets are modest enough that computation is unconstrained by a free runtime, and because the notebook format interleaves code, output and commentary in a single artefact suited to a project whose supervision involves a clinical collaborator who reads results but not code. The cost, that the runtime is ephemeral and package versions are not pinned, is recorded under reproducibility below.

The system is a sequence of stages that each transform data and pass it forward: ingestion, harmonisation of the two external sources onto KATH, feature-set selection, complete-case filtering, encoding, stratified partitioning, model fitting and the three evaluation tasks (Figure 3). The design principle is that the KATH schema is the canonical representation and every other source is translated into it, the pragmatic inverse of the common data model approach in which all sources are mapped onto an external standard [12]. Anchoring on KATH is justified because it is the training institution and the only source carrying the target; the cost is that adding a fourth institution requires a new mapping onto KATH rather than onto a neutral standard, so the layer is extensible in the number of sources but not independent of the training source. Two functions carry the pipeline: an initial run_pipeline establishes the end-to-end path with a simple split and default Random Forest, and run_pipeline_proper supersedes it with stratified three-way partitioning, a selectable model and the fuller metric set; both are retained because the difference between them is itself the baseline result reported in Section 5.1.

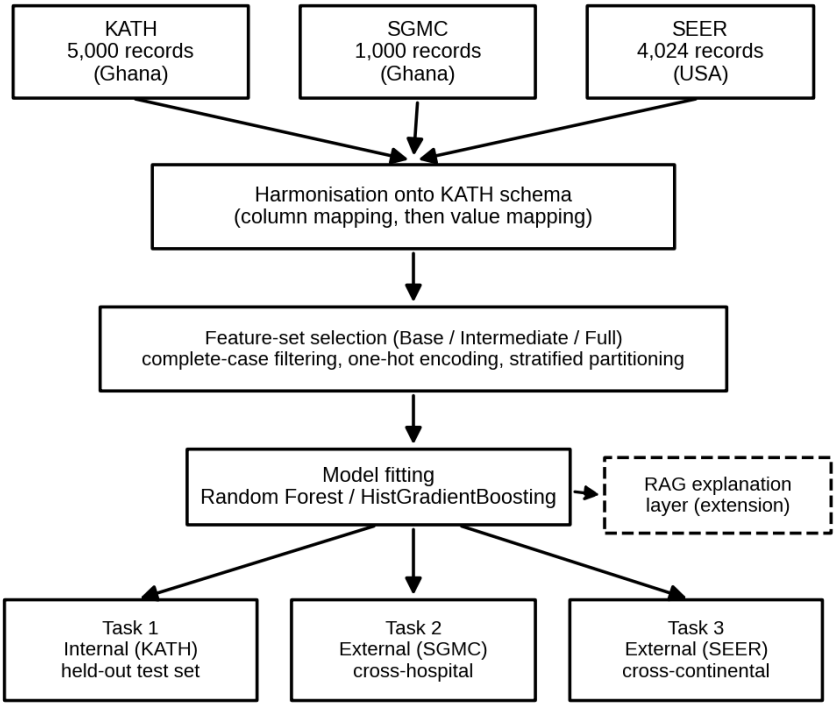


Figure 3: Pipeline architecture, from the three source datasets through harmonisation, feature-set selection, model fitting and the three evaluation tasks, with the RAG layer as an extension on the classifier's output.

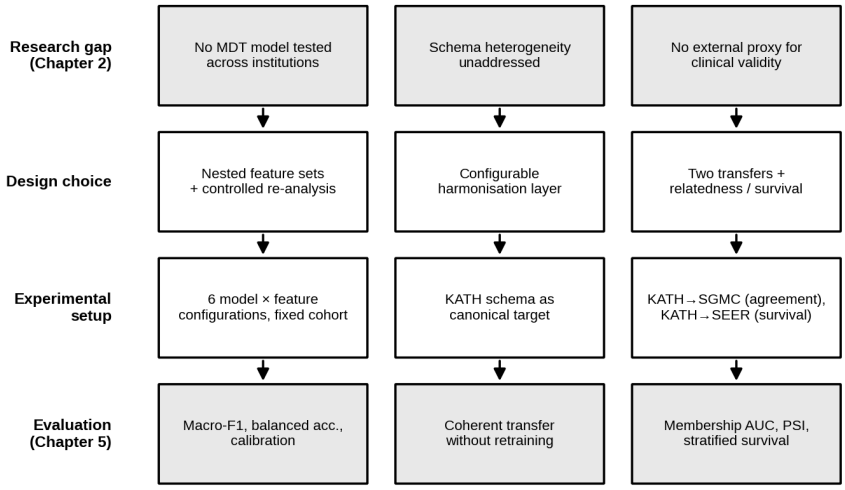


Figure 4: The research-gap-to-evaluation progression. Each design choice traces to a collective limitation identified in Chapter 2 and to the evaluation that tests it in Chapter 5.

4.2 Harmonisation and Preprocessing Implementation

Inspection precedes any transformation, and the ordering is not incidental: the harmonisation layer could only be written after the value vocabularies of each source were enumerated, since the differences are documented nowhere and discoverable only from the data. The SEER extract in particular required inspection because several of its column headers are annotations left by whoever prepared the file rather than variable names; these are excluded by selecting required variables by name rather than position, which also makes the pipeline robust to column reordering.

The column-mapping stage applies a dictionary of twenty-three mappings for SGMC and a smaller set for SEER, which shares only four usable variables, then reduces the source frame to the mapped columns and renames. Reducing to the intersection is deliberate: SGMC records 121 variables, most with no KATH counterpart, and a model can only consume the variables it was fitted with, so the mapped intersection is the operative feature space. The value-mapping stage rewrites the values themselves: HER2 IHC scores to KATH's clinical categories (0 and 1+ to Negative, 2+ to Equivocal, 3+ to Positive), ER and PR Unknown to Pending, stage values prefixed to KATH's form with Unknown mapped to null so it is removed by filtering rather than learned as a class, the T category's leading clinical c stripped, and SEER's substages IIA through IIIC collapsed onto Stage II and Stage III. Value counts are printed before and after every mapping, which is what makes an error visible: an unmapped value silently becomes NaN, so a row that disappears from the harmonised counts is the only evidence that a mapping was incomplete. One consequence is recorded as a genuine loss rather than a format change: KATH distinguishes a Low positive ER category SGMC does not record, and SEER after harmonisation contains only Stage II and Stage III, so the model's exposure to Stage I and Stage IV has no analogue on the receiving data.

Complete-case filtering is applied after feature selection, so the surviving cohort depends on the feature set: 4,923 of 5,000 records survive on the base set, 4,671 on the intermediate and 2,911 on the full, because a row need only be missing one of thirty-two variables to be removed. Categorical variables are one-hot encoded with drop_first, so a variable with k levels yields k minus one indicator columns and the omitted level is represented by zeros, removing the exact collinearity that would otherwise exist. The encoded width grows from 12 columns on the base set to 36 on the intermediate and 51 on the full. For the cross-continental task, encoding KATH and SEER separately would produce different column sets whenever a category present in one is absent from the other, so the two frames are concatenated with source keys, encoded together and split apart again by key, guaranteeing an identical column space by construction, one

of the least obvious failure modes in cross-institutional deployment. The 70/15/15 split is achieved through two stratified train_test_split calls with the seed fixed at 42; a safe_stratify helper returns the target vector when the minimum class count is at least two and None otherwise, causing a fallback to unstratified sampling rather than failure, invoked only when a class is too small to be represented in all three partitions in any case.

4.3 Experimental Setup and Evaluation Components

Models. The Random Forest baseline uses 100 trees at default settings in the initial pipeline; the tuned variant uses 400 trees with min_samples_leaf of 2, the first reducing ensemble variance and the second preventing the tree from isolating single patients, a direct constraint against memorising rare instances. The gradient boosting classifier is HistGradientBoostingClassifier at default settings, chosen over the exact implementation because it bins continuous features into histograms before searching for splits, which is faster and well suited to data of this size. The seed is fixed at 42 for every model. A lightweight grid search over twelve HistGradientBoostingClassifier configurations (max_iter, learning_rate, max_depth) was conducted using the validation partition, selecting the configuration with the highest validation macro-F1 and evaluating it once on the untouched test set, reported in Section 5.4.

Controlled re-analysis. The controlled design is implemented as a distinct stage so both analyses remain executable from the same notebook. Complete-case filtering is applied once against the full feature set before any configuration is fitted; because the sets are nested this single step yields a cohort valid for all three. The partition is computed once and retained as three index objects, and every configuration selects its own columns and slices with those stored indices. This enforces identical patients across configurations: because the indices are computed before any feature selection and never recomputed, no configuration can influence the membership of its own test set. Recomputing the split inside each configuration, as the original pipeline does, would reintroduce the confound even with the same seed, because stratification would operate over a different set of rows. Both classifiers are fitted across all three feature sets in a nested loop from a single code path, making any preprocessing difference between arms impossible by construction.

Target super-grouping. The mapping from fifteen categories to six is implemented as ordered substring tests on the lowercased label, and the ordering is load-bearing: the radiotherapy test excludes labels containing palliative so palliative radiotherapy is not captured as curative, the endocrine test precedes surgery, and the chemotherapy test precedes both. A reordering would silently reassign patients and change the reported accuracy without any error. The final clause is a

catch-all returning Further-workup, which is a residual rather than a treatment decision. The resulting distribution over KATH is Systemic-Chemo 1,772, Surgery 1,276, Palliative/Supportive 1,007, Further-workup 469, Radiotherapy 346 and Endocrine 130.

Cross-hospital agreement checker. Because SGMC records its decision across five fields, a rule-based function determines agreement between the model's single prediction and SGMC's combination, each rule encoding a clinical equivalence. Two properties bound what its output can mean. It is conservative where a rule is simply incomplete (understating agreement) and lenient where it tests consistency rather than identity (overstating it); more importantly it inherits SGMC's base rates, where chemotherapy is recommended for 858 of 1,000 patients and radiotherapy for 796, so a rule conditioned on either is satisfied for most patients regardless of prediction. A base-rate baseline is therefore computed as a designed component of the instrument rather than a post-hoc check: each constant predictor, and a predictor drawing at random from the training prior, is scored through the identical rule set, and the model's rate is reported only as uplift against that reference. The requirement follows directly from the external-validation literature, which documents the uncritical acceptance of unbenchmarked external figures [19,20].

Cross-continental survival analysis. SEER predictions are simplified to three groups by substring match (Chemotherapy, Surgery-first, Other) and joined to survival months and vital status; mean survival and proportion alive are computed per group and a chi-square test of independence applied. The test establishes association only: the predicted groups differ systematically in stage, which independently determines survival, so the association is confounded by construction, and Section 5.5.3 sets out why it cannot be read as validation of clinical correctness.

Calibration, relatedness and shift. The multi-class Brier score and expected calibration error are implemented directly, since neither is available in multi-class form in the dependencies; the class order is read from the fitted model rather than assumed, a small point with a large failure mode, since assuming alphabetical order would misalign the columns and produce a plausible but meaningless score. Empty confidence bins are skipped. The membership model is a binary classification of provenance, fitted on the shared variables encoded together and scored by five-fold cross-validated AUC with out-of-fold predictions so no row scores itself [26]. The Population Stability Index is computed per variable with development-referenced quantile bins for continuous variables and the union of categories for categorical ones, with proportions floored before the logarithm so an empty bin cannot produce an infinite contribution. Impurity-based importances

are reported as a diagnostic but not relied upon, being biased toward continuous and high-cardinality variables such as age.

Reproducibility and delivery. Reproducibility rests on a fixed random seed across every stochastic operation, an explicit inspectable mapping dictionary rather than ad-hoc cleaning, and retention of printed outputs so reported figures trace to the cell that produced them. Results tables are exported to Excel. Two weaknesses are recorded rather than glossed: package versions are not pinned, so a future scikit-learn release could alter a default; and an early implementation constructed summary tables as literal values that drifted out of step with the live output, which is the discrepancy Section 5.1 documents. The delivered notebook constructs tables directly from the result objects, so they cannot diverge.

Table 4.1: Implementation checklist, recording that the required programming and experimental elements were considered and addressed.

Element	How addressed
Reproducible randomness	Seed fixed at 42 across split, bootstrap, subsampling
Deterministic partition	Indices computed once, reused by index across configurations
Missing data	Listwise deletion; counts reported per feature set (Table 5.1)
Encoding integrity	Joint encoding for external transfer; drop_first for collinearity
Stratification safeguard	safe_stratify fallback when a class has fewer than two members
Harmonisation verification	Value counts printed before and after each mapping
Metric correctness	Class order read from fitted model; empty bins skipped
Baseline benchmarking	Constant and prior predictors scored through the same rules
Figure traceability	Tables built from result objects, not literals; outputs retained
Known weaknesses logged	Unpinned versions and under-used validation set recorded

4.4 Retrieval-Augmented Explanation Layer Implementation

The guideline PDF is parsed page by page and split into overlapping chunks of about 180 words with a 40-word overlap, so a recommendation spanning a boundary is not truncated. Boilerplate chunks (copyright footer, page numbering) are filtered by discounting a fixed list of boilerplate terms before applying a minimum real-content threshold, which prevents the retriever returning a near-empty chunk. The guideline produced 232 retrievable chunks from 108 pages after filtering. Each chunk is embedded with a small sentence-transformer (all-MiniLM-L6-v2) [50], and a query is built for each patient by

concatenating the predicted category with the patient's key characteristics (age, stage, ER, PR, HER2). The five highest-similarity chunks are passed to the generation model, Qwen2.5-1.5B-Instruct [40], which is prompted with the prediction, the patient's characteristics and the retrieved passages, and instructed to explain using only those passages, cite page numbers, and state explicitly if the passages do not support the recommendation rather than construct a justification. Both stages run on GPU. Figure 5 shows the data flow.

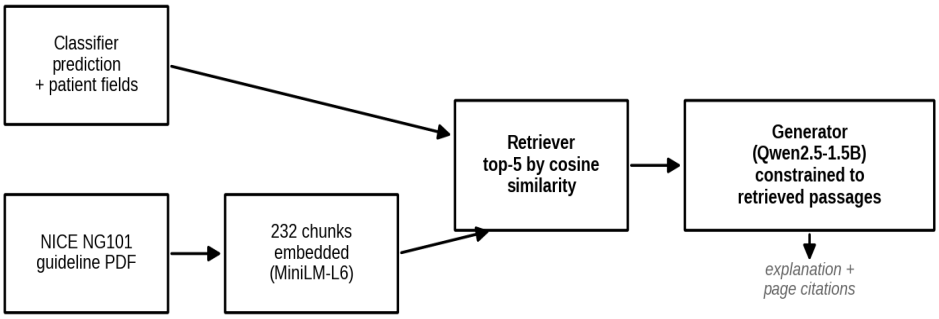


Figure 5: The retrieval-augmented explanation layer. The classifier remains the decision-maker; the generator is constrained to the retrieved guideline passages.

An early prompt permitted freer reasoning and occasionally contradicted the patient's recorded values (describing a HER2-negative patient as positive). A revised prompt forbade inventing any patient detail, forbade citing an unretrieved page, and instructed the model to restate the patient's values exactly. This improved weakly grounded explanations but did not eliminate fabrication of absent patient history, reported as a persistent limitation of the small model rather than a prompt-engineering failure. An automated consistency check comparing each stated receptor status against the true value is retained because it flags some genuine contradictions, but produces false positives when several receptors are discussed in one sentence, so it is reported as a partial safeguard and the results rely on manual review.

The failure the design guards against is named in two literatures that sharpen each other. In natural language generation, hallucination is distinguished into intrinsic (the output contradicts its own source) and extrinsic (the output states something the source can neither confirm nor contradict) [39]; a generated explanation stating a receptor status opposite to the recorded value is intrinsic and checkable against the prompt, while an invented course of prior treatment is extrinsic and requires manual review to detect, which is why the consistency check, targeting only the intrinsic case, is insufficient. In clinical informatics, the corresponding risk is automation bias, the tendency to over-rely on a fluent, confident explanation in proportion to its presentation rather

than its evidential support [43]; the explicit prompt rule that the explanation must never imply the retrieved text validates the recommendation for the specific patient is designed to reduce this at the point of generation. Prior work shows that requiring a model to separate a claim from its verification reduces hallucination [42], and the redesign in Section 5.7 applies that principle to this clinical explanation task. The concordance-faithfulness distinction follows established RAG-evaluation practice, which also observes that faithfulness is harder to automate reliably than retrieval quality, since it compares free text against free text rather than against a fixed reference [41]. The layer is exposed as a single callable function accepting a patient description and returning the prediction with the generated explanation, the interface used to produce the worked examples in Section 5.7.

Chapter Summary. This chapter documented how the methodology of Chapter 3 was implemented. The pipeline architecture anchors on the KATH schema as the canonical representation, with SGMC and SEER translated into it through the harmonisation layer described in Section 4.2. Section 4.3 detailed the experimental setup: model configurations, the controlled re-analysis enforcing identical patients and partitions across feature sets, the target super-grouping logic, and the cross-hospital and cross-continental evaluation instruments, including their acknowledged limitations. Section 4.4 implemented the retrieval-augmented explanation layer, from guideline chunking through retrieval and generation. An implementation checklist (Table 4.1) confirmed that the required programming and experimental elements were addressed, providing the foundation for the results reported in Chapter 5.

5 Results and Evaluation

All figures in this chapter are values computed by the delivered notebook at its most recent execution. This is stated explicitly because an earlier version contained summary tables written as literal values that had diverged from the code's live output, recording for example a base test accuracy of 51.96% against the 57.10% actually computed; these have been removed, on the principle that only a figure a reader can regenerate from the code is defensible. Two classes of result are distinguished throughout: the original pipeline results, which vary feature set and model together, and results from the controlled re-analysis, which hold the cohort and classifier fixed so a single factor varies at a time. Both are reported, because the contrast between them is itself a finding about experimental design.

5.1 Participant Flow and Baseline

Table 5.1 reports the flow of records through the pipeline, following TRIPOD [23,30]. It makes visible a property the per-run counts did not: because complete-case filtering follows feature selection, each feature set is evaluated on a different cohort. The attrition is heavily concentrated. The base set loses 77 records, all to missing ER status; the full set loses 2,089, a little over 40%, dominated by Mammogram_BIRADS (1,150 missing) and Ki67_Percent (988 missing). This matters more than the totals: the full feature set is not a richer description of the same patients but a description of the subset who received a mammogram report and a Ki67 assay, and whether that subset is representative is answered in Section 5.2.

Table 5.1: Participant flow through the KATH pipeline by feature set. Exclusions are records with a missing value among the selected variables only.

Feature set	Vars	Loaded	Excluded	Cohort	Train	Val/Test	Retained
Base	5	5,000	77	4,923	3,447	737 / 739	98.5%
Intermediate	15	5,000	329	4,671	3,271	699 / 701	93.4%
Full	32	5,000	2,089	2,911	2,038	436 / 437	58.2%

The initial pipeline, using the five base variables, an unstratified 80/20 split and a default Random Forest, achieved 54.31% accuracy on the fifteen-class target. Its per-class report is more informative than the headline: neoadjuvant chemotherapy (311 of 985 test cases) achieved an F1 of 0.78 and repeat biopsy/pathology review 0.82, while adjuvant endocrine therapy, adjuvant trastuzumab-based therapy, clinical trial consideration, further imaging and genetic counselling each achieved

exactly 0.00, and best supportive care 0.03. The macro-averaged F1 was 0.29 against a weighted average of 0.51. The gap between the two averages is the finding, not the accuracy: it establishes that the baseline succeeded on frequent categories and failed almost completely on the rest, with five of fifteen categories never predicted correctly. This is the failure mode Section 2.6 identified as concealed by accuracy, and the pattern reported in the most comparable prior study [3]. It also shows why the refinements were necessary rather than cosmetic: the categories the baseline cannot recover, including genetic counselling and clinical trial consideration, are among those where a prompt from a decision-support tool would have most value.

5.2 Task 1: Fifteen-Class Prediction Under a Controlled Design

The original Task 1 analysis varied two factors at once, since model type and cohort both varied with feature set. A controlled re-analysis was therefore conducted, following the requirement that a comparison hold the design fixed and vary one factor at a time [24]. A common cohort of 2,911 records complete on all thirty-two variables exists because the sets are nested; the partition was computed once and the same identifiers reused for every configuration, giving 2,038 training, 436 validation and 437 test patients identical in every row, with both classifiers run across all three feature sets. Table 5.2 reports the result, and the first finding is the principal empirical result of this dissertation.

Table 5.2: Controlled fifteen-class comparison. The 2,911-record cohort is identical across all six configurations, scored on the same 437-patient test set, so feature richness and model type vary independently.

Model	Feature set	Vars	Test acc.	Macro-F1	Bal. acc.
Random Forest	Base	5	56.06%	0.239	25.85%
Random Forest	Intermediate	15	59.95%	0.211	23.55%
Random Forest	Full	32	69.79%	0.356	36.33%
HistGradientBoosting	Base	5	50.34%	0.227	23.19%
HistGradientBoosting	Intermediate	15	54.92%	0.241	25.16%
HistGradientBoosting	**Full**	**32**	**75.29%**	**0.566**	**56.52%**

Feature richness dominates model choice. Within the Random Forest block, moving from five to thirty-two variables raises accuracy by 13.73 points, from 56.06% to 69.79%; within the gradient boosting block the same move raises it by 24.95 points, from 50.34% to 75.29%. Across the model blocks, by contrast, the absolute difference attributable to the classifier is 5.72 points at the base set, 5.03 at the intermediate and 5.50 at the full. The effect of feature richness is therefore between 2.4 and 4.5 times the effect of the algorithm, measured on identical patients with an identical partition. This is the claim stated in Objective 1, now supported by a design that removes

the alternative explanations structurally rather than merely asserting them unlikely.

The cohort confound proved immaterial. This was the more serious threat, because the 2,911 complete-case patients are not a random sample (58.2% retention against 98.5% for the base set). The concern was that they might also be systematically easier to predict. The controlled result settles this: the base configuration scores 57.10% on its own 4,923-record cohort and 56.06% on the restricted 2,911-record cohort, a difference of roughly one point. Restricting to complete cases does not make the problem easier, so the improvement at the full feature set is attributable to the variables rather than to the patients they describe.

The intermediate anomaly was an artefact. The original analysis reported the intermediate set performing worse than the base set (52.35% against 57.10%). Under the controlled design the anomaly disappears: accuracy rises monotonically with feature count for both classifiers. The apparent reversal was produced by the simultaneous change of classifier between the base and intermediate rows, not by any property of the variables. A small non-monotonicity survives in the Random Forest macro-F1 column (0.239 to 0.211 to 0.356), indicating the ten additional intermediate variables improve frequent categories while slightly diluting rare ones, an effect that does not survive at the full set. Finally, the crossover in which the Random Forest leads on the base and intermediate sets but gradient boosting leads on the full set is consistent with the account in Section 3.6: boosting is more expressive but that expressiveness requires signal to exploit, and at five variables encoded into nine columns there is little to find. The original a priori assignment therefore turns out to select the stronger model in each case, retrospectively justified as a modelling choice even though it confounded the original comparison.

5.3 Task 1b: Clinical Super-Groups and Per-Class Behaviour

Table 5.3 reports the same pipeline predicting the six clinical super-groups alongside the fifteen-class figures, on the original per-feature-set cohorts. Grouping raises accuracy by roughly 11 to 20 points across configurations, and the full feature set reaches 86.73% with a macro-F1 of 0.742. The interpretation requires care. Part of the improvement is arithmetic: collapsing fifteen categories into six redefines every confusion internal to a group as a correct answer, so the six-group figure cannot be compared against fifteen-class results elsewhere. The substantive claim is narrower and better supported. The macro-F1 rises from 0.566 to 0.742, and because macro-F1 weights every class equally, this indicates improvement distributed across classes rather than concentrated in the majority. More importantly, the grouped target is the better-posed problem: the fifteen-class task asks the model to distinguish decisions that depend on

information the record does not contain, since whether chemotherapy precedes or follows surgery turns on operability judgements made in the meeting, whereas the grouped task asks which pathway a patient enters, which the recorded data can answer.

Table 5.3: Fifteen-class versus six-group test accuracy on the KATH held-out test set. The two targets are different prediction problems; the columns are not comparable to one another.

Feature set	15-class acc.	6-group acc.	6-group macro-F1
Base	57.10%	75.51%	0.554
Intermediate	52.35%	72.04%	0.465
Full	75.29%	86.73%	0.742

Table 5.4 and the confusion matrix (Figure 6) resolve the aggregate into clinically interpretable behaviour for the strongest configuration, gradient boosting on the full feature set. Radiotherapy was recovered for all 30 of its cases and Palliative/Supportive for 95 of 98. Surgery was correct for 101 of 112, with 11 predicted as chemotherapy, and Systemic-Chemo for 142 of 164, with 14 predicted as surgery. Endocrine was correct for 11 of 13. Further-workup was correct for 0 of 20.

Table 5.4: Per-class performance of the grouped Full model on the KATH held-out test set. Overall accuracy 86.73%.

Group	Precision	Recall	F1	Support
Endocrine	0.917	0.846	0.880	13
Further-workup	0.000	0.000	0.000	20
Palliative/Supportive	0.941	0.969	0.955	98
Radiotherapy	0.789	1.000	0.882	30
Surgery	0.849	0.902	0.874	112
Systemic-Chemo	0.861	0.866	0.863	164
Macro average	0.726	0.764	0.742	437
Weighted average	0.833	0.867	0.849	437

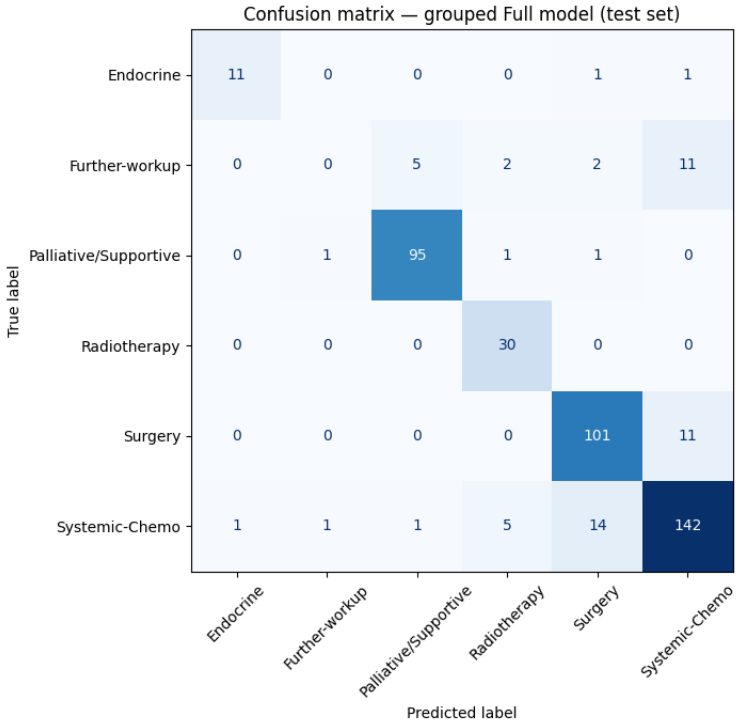


Figure 6: Confusion matrix for the grouped Full model, 437 patients. Rows are the true recommendation, columns the prediction.

The surgery and chemotherapy confusions are the most defensible errors: they are bidirectional and concentrated between two pathways whose boundary is the genuinely contested question in operable breast cancer, whether a patient proceeds to surgery or receives neoadjuvant systemic therapy first. A model that errs at exactly this boundary is failing where human consensus is weakest rather than making arbitrary mistakes. This is qualitatively different from confusing curative and palliative pathways, which the model does not do: Palliative/Supportive is separated at 0.955 F1, and the distinction it is most reliable about is the one carrying the gravest consequence if wrong.

Performance also varies materially by disease stage. Accuracy is lowest for Stage II (75.5%, n=139) and highest for Stage I and Stage IV (94.7% and 94.4%). The Stage IV result is worth reading against its macro-F1 of 0.486, the lowest of any stage group despite the highest accuracy, illustrating exactly the gap between the two metrics that motivates reporting both throughout this evaluation. The Stage II result is consistent with the confusion pattern in Section 5.3: it is the stage at which the surgery-versus-chemotherapy decision is genuinely contested, and the model's errors concentrate at the same clinical boundary human consensus finds hardest.

Table 5.5: Test-set performance by AJCC stage group, grouped Full model.

Stage	n	Accuracy	Macro-F1
Stage I	75	94.7%	0.602
Stage II	139	75.5%	0.629
Stage III	133	88.7%	0.746
Stage IV	90	94.4%	0.486

The Further-workup result is the most important negative finding, and it was tested directly rather than asserted. Section 4.3 argued that this group follows from the target construction: it is not a treatment decision but the residual clause of the grouping function, collecting repeat biopsy, further imaging, genetic counselling and clinical trial consideration, four decisions sharing no clinical logic at the level of the recorded variables. That argument predicts a falsifiable outcome: if the group shares no learnable medical reason across its members, no treatment can recover it. The falsification test was performed. Class weighting was applied to the grouped full configuration, raising the effective cost of a Further-workup error roughly eightfold. Every other class responded, with balanced accuracy rising from 72.72% to 74.55%, but Further-workup remained at precision, recall and F1 of 0.000 across all 20 test cases. The same intervention raised fifteen-class macro-F1 from 0.356 to 0.544, so the treatment is demonstrably capable of rescuing rare but learnable categories. It rescued none of these. The conclusion is therefore evidential rather than interpretive: Further-workup is an unlearnable class, and its unlearnability is a property of the record rather than of the classifier, because what its members share is that the MDT judged the available information insufficient, and the sufficiency of information is precisely what a structured record does not encode. The appropriate remedy is a better target definition, in which incompleteness is modelled as a separate binary flag, proposed in Section 6.4.

Permutation importance for the grouped Full model (Figure 7) shows disease stage dominating: the three stage indicators together account for over half of the total importance among the fifteen most influential features, followed by ECOG Performance Status. Receptor-status variables individually rank low, plausibly because their signal is partly absorbed into the composite Molecular_Subtype variable, which does appear among the stronger features. This extends the finding that the model is close to a stage-driven predictor even with the full configuration available.

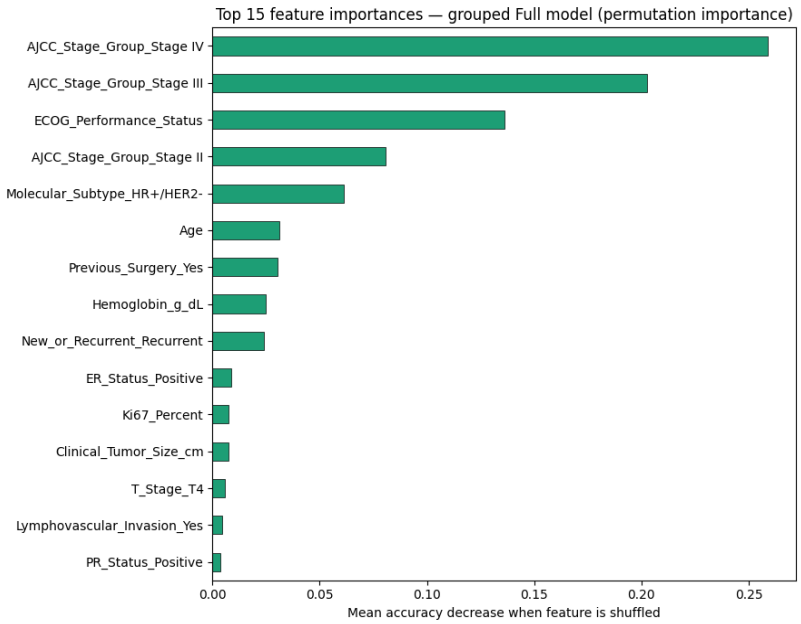


Figure 7: Top fifteen features by permutation importance for the grouped Full model, measured as the mean decrease in accuracy when each feature's values are shuffled.

5.4 Cross-Validation, Imbalance and Calibration

Cross-validation. Five-fold stratified cross-validation of the grouped full configuration returned fold accuracies of 87.31%, 88.49%, 86.25%, 87.11% and 88.83%, a mean of 87.60% with a standard deviation of 0.94%. The 86.73% headline is reported as the conservative point estimate. Two qualifications apply: the cross-validation uses the whole 2,911-record cohort rather than the training partition alone, so its mean is a second estimate over overlapping data, and it remains internal, saying nothing about the distribution shift Sections 5.5 imposes.

Hyperparameter search. A search over twelve configurations, selected by validation macro-F1, did not improve on the untouched default: the tuned configuration (200 iterations, learning rate 0.2, no depth limit) reached 86.04% test accuracy and macro-F1 0.734, against 86.73% and 0.742 for the default. The top five configurations were separated by less than 0.01 macro-F1 on validation, indicating the search could not reliably discriminate between them at this sample size; the default configuration is retained.

Imbalance treatments. Two treatments were evaluated on the controlled cohort with the Random Forest on the full feature set (Table 5.6). Class weighting improves every metric simultaneously: macro-F1 rises from 0.356 to 0.544 (an improvement of 53%), balanced accuracy from 36.33% to 53.59%, and accuracy rises rather than falls, from 69.79% to 72.77%. This is worth remarking on because the expected behaviour is a trade: rebalancing usually recovers rare classes at the cost of frequent ones. That no such trade is observed indicates the

unweighted model was not near a genuine frontier; it was neglecting rare classes without any compensating benefit. SMOTE improves on the baseline but is outperformed by simple class weighting on all three metrics, consistent with the finding that no resampling technique dominates [22] and suggesting that interpolating synthetic patients adds little where minority classes are separated by categorical clinical distinctions rather than by position in a continuous space. Two observations qualify the result: SMOTE balanced every class up to 641 instances (8,974 training instances total) across fourteen classes, not fifteen, so one category was absent from the training partition entirely despite stratification; and the same treatment on the six-group target produced no improvement, because there every learnable class is already handled well and the only failing class is the unlearnable Further-workup residual, so there is no headroom to exploit.

Table 5.6: Imbalance treatments on the controlled cohort, fifteen-class target, Random Forest, full feature set.

Treatment	Test acc.	Macro-F1	Bal. acc.
Untouched	69.79%	0.356	36.33%
Class-weighted	72.77%	0.544	53.59%
SMOTE	70.02%	0.511	50.75%

Calibration. Every metric so far measures discrimination; calibration is co-equal [24], because a model whose stated confidence does not correspond to its observed accuracy is not merely inaccurate but misleading, and the intended tool would surface a confidence alongside each recommendation. The grouped full model returns a multi-class Brier score of 0.226 and an expected calibration error of 0.092 (Figure 8). The distribution of confidence is the more revealing result: of 437 predictions, 383 (87.6%) fall in the highest confidence bin, with a mean stated confidence of 0.993 against an observed accuracy of 0.906. The model is therefore not mildly overconfident on a handful of difficult cases; it declares near-certainty on the overwhelming majority of patients and is wrong in roughly one in ten. The remaining 54 predictions are spread thinly across lower bins, too few to support inference. This is a substantive limitation for deployment, invisible in any accuracy figure, with a plausible mechanism (a tree ensemble trained without calibration on a dominant-pathway target concentrates probability mass) and a standard, inexpensive remedy proposed in Section 6.4.

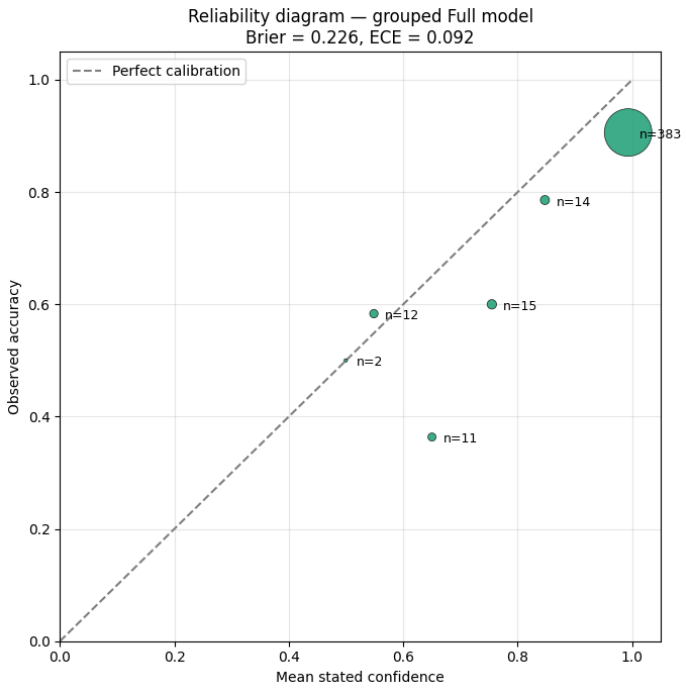


Figure 8: Reliability diagram for the grouped Full model. The diagonal represents perfect calibration; point size reflects the number of predictions in each confidence bin.

5.5 External Evaluation

5.5.1 Population relatedness and residual shift

An external result is uninterpretable without a measure of how far the receiving population sits from the development one, since a drop caused by case-mix is a different finding from one caused by model failure [26]. A membership model was fitted for each transfer. The KATH to SGMC membership AUC is 0.655 and the KATH to SEER AUC is 0.857: the two transfers are not two attempts at the same test but two measured points on a relatedness scale, SGMC moderately related so some loss is expected from case-mix alone, SEER weakly related so transfer there is closer to extrapolation. The Population Stability Index, computed per variable after harmonisation to separate the representational component of shift from the covariate component [27,28], is the important diagnostic. For KATH to SGMC, HER2 status (0.364) and ER status (0.344) exceed the threshold for major shift, stage is moderate (0.219), and age (0.047) and PR status (0.084) are negligible. The receptor-status result matters: these variables are syntactically aligned by the value mapping and remain substantively different, which is a real difference in the referred population rather than a defect in the mapping. For KATH to SEER all four shared variables exceed the threshold (age 1.362, ER 1.227, PR 0.852, stage 5.017). The stage figure should not be read as a routine shift measurement: SEER after harmonisation contains only Stage II and

Stage III, so two of KATH's four stage categories are empty there, and the index records structural absence rather than drift.

5.5.2 Cross-hospital transfer to SGMC

The model trained on KATH's 3,447-patient training partition was applied without retraining to 992 harmonised SGMC patients. The predictions were distributed across nine of the fifteen categories, led by neoadjuvant chemotherapy (434), wide local excision (257) and mastectomy (122). The engineering result this supports is real and is the contribution claimed for Objective 2: a model trained on one hospital's schema produced clinically coherent, distributed predictions on another hospital's data after nothing more than a mapping dictionary, with no retraining and no redesign. The prediction distribution is not degenerate, which is the failure that unmapped value vocabularies would have produced.

The agreement figure itself does not survive scrutiny and is withdrawn. The rule-based check returned 768 agreements of 992 (77.4%), but trivial predictors scored through the identical rules expose the instrument: a constant predictor that always outputs radiotherapy achieves 80.2%, exceeding the model by 2.8 points, and constant neoadjuvant and adjuvant chemotherapy predictors each achieve 78.4%. The model is thus not distinguishable from, and on this evidence marginally worse than, the constant radiotherapy predictor. The correct reading is that the instrument failed, not the model: the rules test consistency with SGMC's fields, and those fields are dominated by near-universal values (radiotherapy recommended for 796 of 1,000 patients, chemotherapy for 858), so a rule conditioned on either is satisfied for most patients regardless of prediction. An agreement measure built on such rules cannot discriminate a good model from a constant, and reporting 77.4% without this comparison would have implied a validation the design cannot deliver, exactly the hazard that motivates benchmarking external figures [19,20]. No accuracy is available for the SGMC transfer, and Section 6.4 sets out what would be required to obtain one.

A further check tests the harmonisation layer directly. Two models were trained on SGMC's own data for a target it records directly (mdt_chemo_recommended), one on native unharmonised values and one on the same patients after harmonisation onto KATH's schema. On a single split the native version scored a higher macro-F1 (0.729) than the harmonised (0.683), consistent with the loss of granularity where SGMC's HER2 categories 0 and 1+ both map to Negative. Repeating across five random splits corrected this: mean macro-F1 was 0.643 native (SD 0.052) against 0.671 harmonised (SD 0.026), a difference smaller than the run-to-run variation either shows, with the harmonised encoding the more stable. The single-split result was not representative, and across five splits harmonisation shows no

consistent cost to predictive power for this target, a direct if narrow answer to whether the effect of schema change is minimal.

5.5.3 Cross-continental transfer to SEER

The third task applied a KATH-trained Random Forest to 4,024 SEER records using the four shared variables (age, stage, ER, PR). HER2, central to treatment selection, is absent from SEER, so this model is necessarily weaker than the full configuration; on the KATH partitions it achieved 54.00% validation and 54.26% test accuracy, which bounds what its SEER predictions can mean. Predictions were distributed across six categories, giving 2,421 patients in a chemotherapy pathway, 1,566 in a surgery-first pathway and 37 in other categories. Patients routed to surgery had a mean survival of 73.6 months with 91.7% alive; patients routed to chemotherapy had 69.9 months with 80.3% alive. A chi-square test of independence returned a statistic of 97.68 with p = 6.15 x 10 to the minus 22.

That test is confounded by disease stage and cannot be interpreted as reported. The model routes patients on stage, because stage is one of only four available variables and the dominant determinant of the KATH decision, and stage also independently determines survival, so the predicted groups differ systematically in stage before any question of recommendation quality arises. Cross-tabulating stage against predicted group shows 100% of Stage III patients routed to chemotherapy and none to surgery, while Stage II patients divide 64.4% to surgery and 35.6% to chemotherapy: on this feature set the model is close to a stage lookup table, and the Stage III stratum contains no surgery patients and cannot be tested. The analysis was therefore repeated within stage strata. Within Stage II, where both pathways are represented, the association survives: 1,566 patients routed to surgery had 73.6 months and 91.7% alive, against 867 routed to chemotherapy with 73.1 months and 88.5% alive, giving a chi-square of 6.44 and p = 0.011. The honest reading is weaker than the unstratified figure suggests and stronger than nothing: the association survives adjustment for the dominant confounder, so the routing carries some information about survival beyond stage alone, but the chi-square falls from 97.68 to 6.44, a reduction of roughly 93%, and the surviving difference is clinically slight (half a month of mean survival, 3.2 points in proportion alive) reaching significance largely because the stratum contains 2,433 patients. The residual remains confounded by age and receptor status, no causal reading is available because SEER does not record treatment received, and the claim in the original objectives that survival serves as an objective proxy for clinical validity is not supported and is withdrawn. What survives is the defensible finding that the harmonisation layer carried the model across a continental boundary and a different coding standard and produced a stage-coherent distribution rather than collapsing.

5.6 Threats to Validity and Summary of Findings

Internal validity. The two confounds identified in the original analysis were resolved rather than acknowledged: the model confound by running both classifiers across all three feature sets, and the cohort confound by evaluating every configuration on a common 2,911-record cohort with an identical partition, which changes base-set accuracy by about one point. The residual limitation is that the controlled analysis necessarily discards the 2,089 records excluded by complete-case filtering, so it speaks only to fully documented patients. Hyperparameters were not searched, and the validation partition, though correctly constructed, was not used for selection, so the figures are attainable rather than optimised. No calibration was applied, and Section 5.4 quantifies the consequence.

Construct validity. The target is the recommendation KATH's MDT recorded, not the clinically optimal decision, so a high accuracy denotes fidelity to KATH's practice, including any systematic bias in it, rather than clinical correctness, and this qualification propagates to every result. The six-group target contains the Further-workup residual, now shown experimentally not to be a coherent construct. The SGMC agreement instrument measures rule-based consistency rather than accuracy and is indiscriminate; its output is withdrawn. The SEER survival analysis is confounded by stage, and the residual association after stratification remains confounded by the model's other predictors.

External validity. Both transfers use a hand-written mapping and both receiving datasets are narrower than the source. The relatedness measurements quantify the distance (AUC 0.655 to SGMC, 0.857 to SEER), so neither transfer is to a comparable population and the SEER transfer in particular is extrapolation. Harmonisation resolves only the representational component of shift; HER2 and ER remain majorly shifted between KATH and SGMC after syntactic alignment, so the transfer results establish that the representational barrier can be removed, not that the populations are exchangeable. Read in this light, the argument that site heterogeneity is a real property of practice [29] suggests the appropriate role for the layer is to enable local validation rather than to deliver a model assumed to transfer unchanged. The stage-collapsing mapping (Section 3.4) is a genuine limitation on this basis: full TNM staging informs management in a way a single collapsed category does not, and the clinical-versus-pathological staging distinction is not captured at all, which would need to be addressed by retaining TNM sub-components as separate fields before a tool of this kind could realistically support clinical decision-making.

Against the research questions, the evidence supports the following. On RQ1, a classifier predicts the fifteen-class recommendation at

75.29% with the full feature set, and performance is governed by data richness rather than algorithm choice, the effect of moving from five to thirty-two variables being between 2.4 and 4.5 times that of changing the classifier on identical patients. On RQ2, the six-group target raises accuracy to 86.73% and macro-F1 to 0.742, stable at 87.60% (SD 0.94%); part of this gain is arithmetic, but the grouped target is the better-posed problem, and one of its classes is demonstrably unlearnable. On RQ3, the layer carried the model to a second hospital and produced coherent predictions on 992 patients without retraining, a genuine engineering result, but the 77.4% agreement figure is withdrawn since constant predictors reach 80.2%. On RQ4, the layer carried the model across continents and produced stage-coherent predictions, but the four-variable model is close to a stage lookup table and the survival association is 93% attributable to stage, so the survival-as-proxy claim is withdrawn. Two findings recur and are the most consequential: the binding constraint on performance is the data rather than the algorithm, and the harmonisation layer addresses only the representational barrier to transfer while the substantive barriers remain. For a system intended where data capture is the binding constraint, the first locates the problem and the second bounds the claim.

5.7 Evaluation of the Retrieval-Augmented Explanation Layer

The layer was evaluated on the 30-patient held-out test set used throughout, with the classifier's own prediction taken as the recommendation to be explained, plus five constructed profiles spanning triple-negative, hormone-receptor-positive, HER2-positive, advanced-stage and mixed-receptor presentations, reported as worked examples rather than an independent sample. Sixteen of thirty predictions retrieved passages containing terms directly associated with the predicted category; the remaining fourteen did not. This roughly even split is consistent with the established limitation that the classifier is trained on five variables and cannot draw on the tumour size, nodal involvement or margin status a guideline recommendation typically depends on: where the prediction is under-determined by its own inputs, retrieval correctly finds no clean textual match, which is the evaluation surfacing a known limitation rather than a fault introduced by the explanation layer.

Manual review of all thirty explanations identified five recurring failure patterns rather than a single undifferentiated hallucination rate (full detail in Appendix B). The most frequent and informative was fabricated treatment history, in six of thirty explanations and concentrated entirely in one guideline region, the pages on trastuzumab-based adjuvant therapy, which suggested a checkable cause rather than an unlocalised failure. This motivated a testable hypothesis: the pattern arises because the prompt asks the model to explain why the prediction is correct, which invites it to close any gap

between retrieved text and prediction by inventing supporting detail, conflating what the text says with whether it establishes the prediction for this individual.

The prompt was redesigned to keep these two questions structurally separate, requiring five explicitly labelled sections (Patient Characteristics, ML Prediction, Relevant NICE Recommendations, Evidence-Based Interpretation, Limitations), with an instruction that the model must never imply the retrieved text confirms the recommendation for the specific patient and that the Limitations section must state plainly that the guidance does not by itself establish the recommendation, since that requires factors not captured in the five available characteristics. The redesign was evaluated on the same 30 patients and five profiles, holding classifier, retriever and generation model identical so the comparison isolates the prompt change. Table 5.7 and Figure 9 compare the two.

Table 5.7: Failure patterns before and after the prompt redesign. Four of the five patterns were eliminated entirely; the fifth changed behaviour materially.

Pattern	Original (30)	Redesigned (30)
Fabricated treatment history	6 of 30	0 of 30
Self-contradiction on a receptor	2 of 30	0 of 30
Wrong drug class for HER2 status	2 of 30	0 of 30
Unsupported or reversed clinical claim	2 of 30	0 of 30
Reasoning does not support recommendation	3 of 30	4 of 30 (see text)

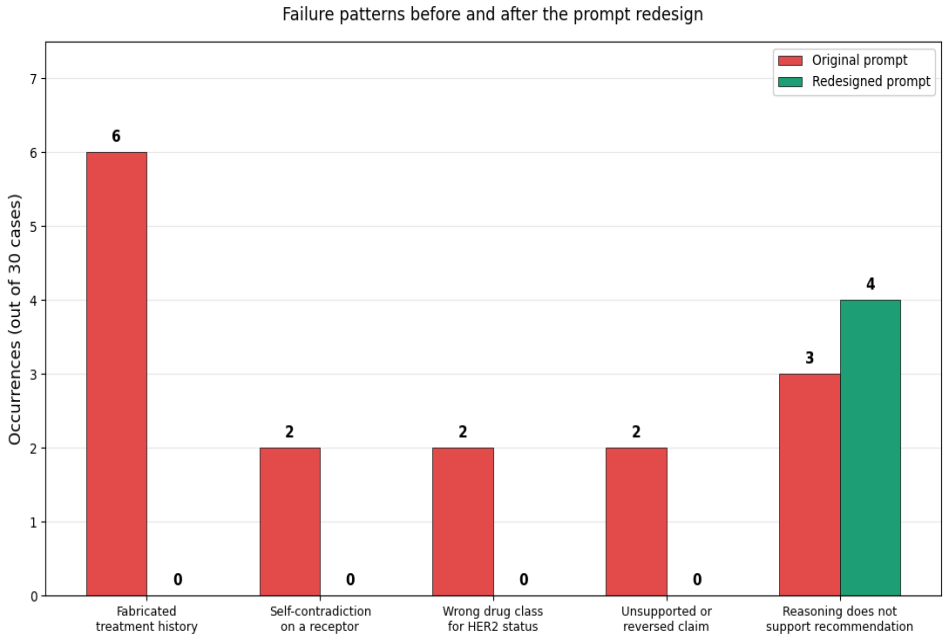


Figure 9: Failure-pattern counts before and after the prompt redesign, by category, corresponding to Table 5.7.

Four of the five patterns do not recur under the redesigned prompt. The fabricated-history pattern, the most frequent and concerning finding, does not appear once: every explanation referencing trastuzumab-based eligibility now frames it as a general guideline condition rather than asserting the patient has received it. The Palliative/Supportive cases (four occurrences, where the retrieved evidence points toward chemotherapy or endocrine therapy while the prediction is Palliative/Supportive) persist but behave differently: every such case now states explicitly in the Limitations section that the retrieved guidance does not confirm palliative care is indicated, the safeguard operating as intended. One such case, an advanced-stage hormone-receptor-positive patient, illustrates the distinction between treatment type and treatment intent: palliative-intent chemotherapy is standard practice, so guideline text discussing chemotherapy does not by itself rule out a palliative recommendation. The consistency check cannot represent this distinction, which is a limitation of the check rather than of the prediction.

Two residual weaknesses remain. A softer echo of the original overreach appears in seven of thirty explanations, the phrase which aligns with the ML prediction stated before the Limitations section correctly qualifies it, with no fabricated detail attached; it indicates the separation of description from validation is not perfectly followed. And worked-example review surfaced a pattern not captured by the five categories, mischaracterisation of the retrieved source text itself: one explanation described a passage on BRCA-mutation-positive disease as though it concerned hormone-receptor-positive disease, which the automated check cannot detect because it compares only against the patient's own values. The single clearest fabrication under the redesigned prompt was an explanation referring to a fifty-year-old male patient; sex is not among the five characteristics used anywhere, so this cannot have come from the input, an unambiguous invention confirming that a model of this size does not eliminate fabrication entirely. Taken together, the redesign is a substantial and measured improvement rather than a resolved problem, and the evaluation process itself, diagnosing a specific failure mechanism, redesigning the prompt to target it, and re-evaluating on the same sample rather than a favourable new one, is offered as the more durable contribution of this component.

Chapter Summary. This chapter reported the dissertation's central evidence. A controlled re-analysis established the principal finding that feature richness governs predictive performance far more than classifier choice, with the six-group task reaching 86.73% accuracy and the Further-workup category shown by a falsification test to be

genuinely unlearnable rather than merely neglected. Cross-hospital transfer to SGMC and cross-continental transfer to SEER both demonstrated that the harmonisation layer carries the model across differing schemas and populations, though the cross-hospital agreement figure and the survival-as-proxy claim were both critically examined and withdrawn once shown to be indiscriminate or confounded. A hyperparameter search and a stage-subgroup breakdown were added to test whether the reported result was sensitive to model tuning and to patient stratification respectively, and the RAG layer's evaluation, including a documented prompt redesign, closed the chapter's account of what this dissertation's evidence does and does not support.

6 Conclusion and Future Work

6.1 Conclusions and Objectives Achieved

This dissertation set out to design, implement and evaluate an end-to-end machine-learning pipeline predicting breast cancer MDT treatment recommendations that remains applicable across institutions recording data differently, a problem left open by prior work confined to single well-resourced institutions. All four objectives were met, though not all with equal strength of evidence.

The three nested feature sets, the two-stage configurable harmonisation layer, and the two external transfers to SGMC and SEER were all delivered, the transfers with the relatedness of each receiving population measured so they could be interpreted rather than merely reported. A documented, reproducible pipeline was delivered, and the retrieval-augmented explanation layer was designed, implemented and evaluated, including a redesign in response to the first evaluation, as a proof-of-concept rather than a clinically validated component.

The principal finding is that predictive performance here is governed principally by the richness of recorded data rather than by the choice of classifier: on identical patients with an identical partition, moving from five to thirty-two variables changed accuracy by between 13.7 and 25.0 percentage points, against roughly 5.5 for the algorithm. The six-group target reached 86.73% accuracy (macro-F1 0.742), stable under cross-validation, though one class, Further-workup, was shown by a falsification test to be unlearnable rather than merely neglected. The transfers established that the harmonisation layer carries a model across markedly different schemas and a continental boundary and produces coherent, stage-appropriate predictions, while the candid analysis showed that the cross-hospital agreement figure and the survival-as-proxy claim could not be sustained and were withdrawn.

6.2 Contributions

The principal contribution is the demonstration that a lightweight, inspectable harmonisation layer is sufficient to carry a supervised MDT recommendation model across institutions with markedly different schemas and value vocabularies, without retraining or redesign. The reviewed literature treats schema heterogeneity either as out of scope [2,3] or as a problem requiring enterprise-scale standardisation [11,12]; this work occupies the space between and shows that for a bounded task the interoperability principle can be retained at a fraction of the cost. A second, empirical contribution is that performance is governed by data richness rather than classifier choice, established under a controlled design, which locates the binding constraint for institutions considering decision support in data capture rather than modelling capability. A third is methodological: reporting the fifteen-

class and six-group targets side by side separates the gain from target design from that of modelling, and the same discipline produced two results a less careful design would have missed and a less candid one suppressed, namely the identification of Further-workup as unlearnable by falsification test and the demonstration that the cross-hospital agreement instrument is indiscriminate. A fourth follows from the RAG extension: an automated faithfulness measure was attempted, found unreliable, and replaced with manual review that identified five distinct failure patterns rather than a single hallucination rate, one concrete enough to guide future correction.

6.3 Limitations

The limitations are set out in full in Section 5.6 and summarised here. Two confounds identified during the work were resolved by the controlled re-analysis; those remaining are as follows. The controlled analysis speaks only to fully documented patients, excluding the 2,089 records removed by complete-case filtering. No calibration was applied, and the model is materially overconfident: 87.6% of predictions are issued at a mean confidence of 0.993 against an observed accuracy of 0.906. The target is the recommendation KATH recorded rather than the optimal decision, so accuracy denotes fidelity to one institution's practice. The cross-hospital agreement rate is withdrawn, since constant predictors reach 80.2% under the same rules, and the SEER survival association is 93% attributable to disease stage. Finally, the system was evaluated only retrospectively; no clinician participated in training, in evaluating the prediction sample as a whole, or in any live decision.

6.4 Future Work

- The following improvements follow directly from the limitations and findings, ordered by the ratio of evidential gain to effort.
- Extend the controlled comparison to incomplete records by evaluating imputation against listwise deletion. The central claim is established among fully documented patients; whether it holds for the 2,089 excluded records is untested, and those are precisely the records a lower-resource institution would contribute.
- Retain TNM sub-components as separate fields rather than collapsing to a single stage category, and distinguish clinical from pathological staging where a source records them, since the current simplification discards clinically material information; it is a low-effort change for KATH and SGMC and would need to precede any claim of clinical utility.
- Obtain a genuine external accuracy for the SGMC transfer through a clinically agreed mapping from SGMC's separate decision fields onto a single recommendation, authored with the

clinical collaborator rather than derived post hoc, since the rule-based agreement instrument cannot discriminate model quality.

- Apply post-hoc probability calibration to correct the near-certain but only nine-in-ten-correct predictions, the defect most likely to erode clinical trust and inexpensive to remedy; a hyperparameter search was found not to help (Section 5.4), so calibration is the remaining low-cost fix.
- Reconstruct the target so Further-workup is a separate binary flag for case incompleteness rather than a treatment class, since it is a statement about the sufficiency of the record rather than a management decision.
- Adopt class weighting as standard for the reported configurations, since it raised fifteen-class macro-F1 from 0.356 to 0.544 with accuracy rising rather than falling.
- Extend the RAG layer by investigating the fabricated-history pattern's association with the trastuzumab guideline region directly, and by evaluating a larger generation model where resources allow, to establish whether persistent fabrication is a function of model scale rather than prompt design.
- Conduct a prospective evaluation with clinicians in the loop, measuring whether the system changes triage time or decision consistency, which is the only design capable of establishing clinical usefulness.

Taken together, extending the controlled comparison to incomplete records and obtaining a genuine external accuracy for the SGMC transfer would materially strengthen the claims already made without collecting new patient data, and represent the most efficient path to a defensible submission.

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Appendix A Full Field and Value Mappings

This appendix expands Table 3.1 with the complete column-name and value-vocabulary mappings the harmonisation layer applies. It contains no information beyond Section 3.4; it is provided as a single reference for readers who want the full mapping without searching the prose.

A.1 Column-name mapping

Clinical concept	KATH column	SGMC column	SEER column
Age	Age	age_years	Age
Stage group	AJCC_Stage_Group	stage_group_derived	6th Stage
Oestrogen receptor	ER_Status	er_status	ER_Status
Progesterone receptor	PR_Status	pr_status	PR_Status
HER2 status	HER2_Status	her2_ihc	Not recorded
Tumour category (T)	T_Category	t_category	T_Category

A.2 Value mapping: HER2 status

KATH records HER2 as a clinical category; SGMC as a raw immunohistochemistry score; SEER not at all. Scores 0 and 1+ both map to Negative, following the interpretation in which a 2+ result is equivocal pending reflex in-situ hybridisation and only a 3+ score is a definitive positive.

SGMC value (raw IHC score)	KATH value (clinical category)
0	Negative
1+	Negative
2+	Equivocal
3+	Positive

A.3 Value mapping: stage group

KATH and SGMC record stage at the main-group level (Stage I to IV); SEER records substages (for example IIA, IIB). The mapping collapses SEER's substages onto KATH's main groups, so IIA and IIB both map to Stage II, and IIIA, IIIB and IIIC all map to Stage III. This is lossy: substage encodes prognostically meaningful information about tumour and nodal burden that is discarded in the collapse. The reverse mapping is not available, since KATH and SGMC never recorded substage, so harmonisation can only level down to the coarsest source.

A.4 Value mapping: T category

T category values are recorded with minor formatting differences; for example, a tumour recorded as T2 at KATH is cT2 at SGMC, where

the leading c denotes a clinical, pre-treatment assessment. The value mapping strips this prefix so the underlying T category is compared on a common basis.

Appendix B First-Pass RAG Evaluation Detail

This appendix gives the full detail of the first-pass manual review referenced in Section 5.7, prior to the prompt redesign. Table B.1 reproduces the failure-pattern breakdown.

Table B.1: Failure patterns identified by manual review of 30 generated explanations. Categories are not mutually exclusive; six explanations exhibited no pattern and were judged fully supported.

Pattern	Occurrences	Example
Fabricated treatment history	6 of 30	States the patient has already received trastuzumab-based therapy, a fact not in the five inputs, citing a technology appraisal never retrieved
Self-contradiction on a receptor	2 of 30	States a receptor positive in one clause and negative in the next, for the same patient
Reasoning does not support recommendation	3 of 30	Argues the profile favours chemotherapy, then concludes with a Palliative/Supportive recommendation without resolving the discrepancy
Wrong drug class for HER2 status	2 of 30	Suggests trastuzumab or pertuzumab, HER2-targeted agents, for a patient recorded as HER2-negative
Unsupported or reversed clinical claim	2 of 30	States a hormone-receptor-positive, HER2-negative profile carries higher recurrence risk than a HER2-positive profile, the reverse of standard understanding

The fabricated-history pattern is the most informative because it is not evenly distributed: all six occurrences cite the same guideline region, the pages on trastuzumab-based adjuvant therapy, suggesting the fabrication is prompted by the phrasing of that specific passage rather than occurring at random, and giving a developer a concrete, checkable target. The first prompt revision measurably improved radiotherapy explanations, which had been self-contradictory on receptor status and now correctly restated the patient's values and cited a real recommendation on partial-breast radiotherapy eligibility with numeric criteria. The same revision did not reduce the fabricated-history pattern, which recurred at a similar rate: prompt wording can correct errors that stem from reasoning poorly over provided text, but cannot prevent a small model from filling a genuine information gap with an invented detail. Two explanations demonstrated the intended safeguard working, stating explicitly that the retrieved passages did not support the recommendation rather than constructing a justification, which shows the model is not incapable of the required behaviour but does not exhibit it reliably at this scale.

Appendix C Worked Examples of the RAG Explanation Layer

This appendix presents two worked examples of the RAG explanation layer's generated output, illustrating the format described in Section 4.4: a 40-year-old patient with metastatic disease predicted Palliative/Supportive, and an 87-year-old patient predicted Endocrine therapy. Both are discussed as evaluation examples in Section 5.7.

```
Patient 12: Palliative/Supportive
**Patient Characteristics:** Age: 40
AJCC_Stage_Group: Stage IV
ER_Status: Positive
PR_Status: Positive
HER2_Status: Negative
ML Prediction: Palliative/Supportive

**Relevant NICE Recommendations:**
The retrieved NICE passages indicate that hormone receptor (HR) positive, HER2
negative breast cancer should be managed through neoadjuvant chemotherapy where
indicated. Specifically, the passage states: "Consider neoadjuvant chemotherapy,
where indicated, for people with ER-positive invasive breast cancer." Additionally,
the guideline recommends olaparib, alone or with endocrine therapy, as an option for
early and locally advanced breast cancer if certain criteria are met. However, the
passage does not specify whether this treatment is appropriate for a patient with HR
positive, HER2 negative stage IV disease based solely on the provided information.

**Evidence-Based Interpretation:**
Based on the retrieved NICE passages, neoadjuvant chemotherapy is considered an
option for patients with HR positive, HER2 negative breast cancer, but its
appropriateness for a patient with HR positive, HER2 negative stage IV disease
cannot be determined without additional information such as tumour grade, nodal
status, and a comprehensive multidisciplinary assessment.

**Limitations:**
The retrieved NICE passages provide guidelines for managing HR positive, HER2
negative breast cancer, but they do not confirm the appropriateness of neoadjuvant
chemotherapy or other treatments for a specific patient like the one described.
Therefore, the decision regarding the use of neoadjuvant chemotherapy or any other
treatment must be made after considering all relevant factors and conducting a
thorough evaluation of the patient's case.
```

```
Patient 27: Endocrine
**Patient Characteristics:** Age: 87
AJCC_Stage_Group: Stage II
ER_Status: Positive
PR_Status: Negative
HER2_Status: Negative
ML Prediction: Endocrine

**Relevant NICE Recommendations:**
The retrieved NICE passages indicate that adjuvant endocrine therapy is recommended
for patients with ER-positive invasive breast cancer. Specifically, the passage
states: "Offer adjuvant endocrine therapy to people with ER-positive invasive breast
cancer." This recommendation applies to both early and locally advanced stages of
breast cancer.

**Evidence-Based Interpretation:**
The retrieved NICE passage supports the recommendation to offer adjuvant endocrine
therapy to patients with ER-positive invasive breast cancer. However, it does not
provide enough information to determine if this recommendation is applicable to a
87-year-old patient with stage II breast cancer, given the lack of additional
context such as tumour grade, nodal status, or a full multidisciplinary assessment.

**Limitations:**
The retrieved guidance does not establish that adjuvant endocrine therapy is
indicated for this specific patient due to the absence of critical patient-specific
factors like tumour grade, nodal status, or a comprehensive multidisciplinary
evaluation.
```